QUANTITATIVE FINANCE RESEARCH PROJECT

MARKET DNA

Detecting Structural Identity Drift Across Financial Assets

A cross-asset framework for measuring changes in factor exposures, market regimes and portfolio risk

Python • R

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Executive Summary

Financial assets are commonly classified using relatively stable labels such as growth equities, financial stocks, energy equities, government bonds or gold. However, the economic behaviour of an asset may change substantially through time even when its conventional classification remains unchanged. This project develops a quantitative framework, called Market DNA, to measure these changes through the evolution of asset sensitivities to major financial-market factors.

The analysis covers six liquid cross-asset instruments — QQQ, XLF, XLE, FEZ, TLT and GLD — over the period from January 2010 to September 2026. For each asset, a rolling multi-factor model estimates time-varying sensitivities to the global equity market, long-term interest rates, credit conditions, the US dollar, oil prices and market volatility. These six rolling exposures form the asset's Financial DNA. Structural changes in this exposure vector are quantified using Mahalanobis distance, first relative to a fixed historical benchmark and subsequently through a dynamic rolling reference framework.

The resulting Dynamic Market Identity Drift Index (DMIDI) measures the average degree of structural displacement across the six assets. The framework is complemented by Drift Breadth, an entropy-based measure that distinguishes concentrated changes in a small number of assets from broad structural shifts affecting the cross-asset system. Principal component and correlation analysis show that structural drift is multidimensional rather than being dominated by a single common component.

The empirical results also provide an important negative finding. Neither the fixed nor the dynamic Market DNA indicators improve out-of-sample forecasts of future market volatility relative to conventional controls dominated by the VIX. Market DNA should therefore not be interpreted as a volatility-prediction model. Evidence for future cross-asset correlation is more encouraging but remains modest: a nonlinear DMIDI specification produces small recurrent out-of-sample improvements across several years, although the pooled improvement is not statistically strong enough to support a claim of forecast superiority.

The framework is subsequently translated into a percentile-based market-regime map and a scenario-conditioned Monte Carlo risk engine. At the end of September 2026, DMIDI stands at approximately 10.71, while Drift Breadth is 0.847. Relative to the 2015–2021 reference distribution, structural drift is elevated but unusually concentrated, with GLD accounting for approximately 44.7% of total asset-level drift. The resulting state is classified as a Concentrated Shift rather than a broad systemic transformation.

Overall, the project suggests that Market DNA is most useful as a framework for detecting, decomposing and interpreting structural changes in financial-market relationships, rather than as a standalone forecasting signal. Its main contribution is to provide a transparent link between evolving factor exposures, cross-asset structural regimes and portfolio-risk scenarios while preserving a clear distinction between descriptive structural information and predictive evidence.

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1. Research Motivation

1.1 Why Financial Identity May Change Over Time

Financial assets are often described through relatively stable labels: technology equities, financial stocks, energy companies, government bonds, gold, or international equities. These classifications are useful, but they do not necessarily imply that the economic behaviour of an asset remains stable through time. The same asset may react very differently to changes in interest rates, credit conditions, the US dollar, oil prices or market volatility depending on the prevailing macro-financial environment.

This project starts from the idea that an asset can be characterised not only by its return or volatility, but also by the combination of sensitivities that links it to major market forces. These sensitivities can be interpreted as a form of **financial DNA**. Rather than treating factor exposures as permanent characteristics, the framework allows them to evolve through time.

Consider, for example, an equity index whose performance has historically been strongly associated with the broad equity market but only weakly related to interest rates. If its rate sensitivity subsequently becomes large and persistent, the asset may still carry the same ticker and belong to the same conventional asset class, while its effective economic behaviour has changed. Similarly, gold may alternate between behaving primarily as a defensive asset, an inflation-sensitive asset, or an asset strongly influenced by the US dollar and interest-rate environment.

Such changes matter because traditional risk measures can describe how much an asset is moving without necessarily explaining how its economic relationships are changing. A rise in volatility captures larger price fluctuations, while a change in factor exposure captures a different dimension: a change in the mechanism through which the asset interacts with the broader financial system.

Market DNA therefore focuses on **structural change rather than on price movements alone**. The central premise is that a financial asset has a time-varying identity represented by its exposure to a set of common market factors, and that sufficiently large changes in this exposure vector can be measured, compared across assets and aggregated into indicators of broader market transformation.

> **Core idea**
> An asset's identity is represented by its time-varying sensitivities to major market factors. Market DNA measures when these relationships move materially away from their recent historical structure.

1.2 Research Question

The project investigates whether changes in the factor structure of financial assets can be identified systematically and whether those changes contain information about broader market conditions.

The analysis is organised around three related questions:

1. Can structural changes in an asset's factor exposures be measured in a consistent way?

2. Are these changes concentrated in individual assets or distributed across the financial system?

3. Do periods of structural identity drift contain information about subsequent market risk, cross-asset correlation or portfolio outcomes?

The initial hypothesis was that large structural drift might anticipate higher future volatility. This hypothesis is explicitly tested rather than assumed. As the empirical analysis develops, the project therefore distinguishes between **measuring structural change**, **explaining where it originates**, and **testing whether it has predictive value**.

1.3 Contribution of the Project

Market DNA combines several established quantitative techniques within a single cross-asset framework. The contribution of the project lies primarily in their integration and interpretation rather than in claiming that any individual statistical method is new.

First, rolling factor models are used to represent the evolving financial DNA of six assets spanning equities, bonds and commodities. Second, multivariate distance measures are applied to quantify how far each asset's current exposure structure has moved from a historical reference. These asset-level measures are then aggregated into a Dynamic Market Identity Drift Index (DMIDI).

The framework is extended with a measure of **drift breadth**, which distinguishes a structural change concentrated in a small number of assets from a broad shift affecting several markets simultaneously. Principal component analysis and cross-asset drift correlations are then used to investigate whether structural change is driven by a common market component or remains largely asset-specific.

The indicators are subsequently subjected to both in-sample and genuine out-of-sample testing. This distinction is important: the objective is not simply to identify statistically interesting historical relationships, but to determine whether the information survives when models are estimated using only data available before the evaluation period.

Finally, the project translates the Market DNA framework into portfolio-risk scenarios through a block-bootstrap Monte Carlo engine. This final layer connects structural regime detection with practical risk measures including Value at Risk, Expected Shortfall, loss probabilities and maximum drawdown.

The resulting framework follows a simple progression: detect structural change, explain its source, test its information content, and simulate its potential portfolio consequences.

2. Data and Factor Architecture

The Market DNA framework requires a cross-asset universe broad enough to capture different economic transmission channels while remaining sufficiently compact for interpretation. The empirical analysis therefore combines six liquid exchange-traded assets representing US equities, financials, energy, European equities, long-duration US government bonds and gold. These assets are analysed against six common market factors designed to capture broad equity conditions, interest rates, credit, the US dollar, oil and market volatility.

2.1 Asset Universe

The analysis focuses on six assets with deliberately different economic characteristics. The objective is not to construct an investable benchmark, but to observe whether structural changes in factor exposures occur simultaneously across markets or remain concentrated in specific asset classes.

The selected universe combines growth-sensitive equities, financial and energy sectors, European equities, duration exposure and a traditional defensive asset. This diversity makes it possible to distinguish between asset-specific identity drift and broader changes in financial-market structure.

Table 1. Cross-Asset Universe Used in the Market DNA Framework

Ticker	Asset / Proxy	Economic Role	Why Included
QQQ	Nasdaq-100 ETF	US growth / technology equities	Captures growth-sensitive equity exposure and changing sensitivity to rates and market risk
XLF	Financial Select Sector ETF	US financial equities	Represents banks and financial institutions with strong links to rates and credit conditions
XLE	Energy Select Sector ETF	US energy equities	Provides exposure to energy-sector dynamics and oil-related shocks
FEZ	Euro Stoxx 50 ETF	Large-cap euro-area equities	Adds international equity exposure and sensitivity to European macro-financial conditions
TLT	20+ Year US Treasury ETF	Long-duration government bonds	Represents duration and interest-rate-sensitive defensive exposure
GLD	Gold ETF	Gold / defensive asset	Captures a non-equity asset influenced by rates, the US dollar and risk sentiment

*Note: **ETFs** are used as liquid market proxies for broad asset and sector exposures. The selected universe is intended for structural cross-asset analysis rather than portfolio optimisation.*

The universe is intentionally heterogeneous. QQQ, XLF, XLE and FEZ provide different forms of equity exposure, while TLT and GLD introduce assets whose economic behaviour may diverge materially from equities. This cross-asset structure is essential for the later construction of Drift Breadth, since a high aggregate drift driven by one asset carries a different interpretation from a simultaneous reconfiguration across equities, bonds and commodities.

2.2 Market Factors

The Financial DNA of each asset is defined through its sensitivity to six common market factors. The factor set is deliberately compact: it is designed to capture major macro-financial transmission channels while preserving economic interpretability and limiting unnecessary model complexity.

The selected factors represent broad equity-market conditions, long-term interest rates, credit conditions, the US dollar, oil prices and market volatility. Together, they provide a common explanatory structure that can be applied consistently across equities, bonds and commodities.

Table 2. Market Factors Used to Define Financial DNA

Factor	Proxy / Construction	Economic Interpretation	Transformation
MKT	VT	Global equity market	Daily log return
RATE	US 10-Year Treasury Yield (DGS10)	Long-term interest-rate conditions	Daily yield change in basis points
CREDIT	HYG − IEF	Credit-risk conditions relative to government bonds	Difference between daily log returns
USD	UUP	US dollar strength	Daily log return
OIL	WTI crude oil futures (CL=F)	Energy and commodity-price conditions	Daily log return
VIX	CBOE Volatility Index (VIXCLS)	Changes in market-implied uncertainty	Daily change in VIX level

Note: MKT, CREDIT, USD and OIL are return-based factors. RATE and VIX are expressed as changes in levels. The VIX level itself is retained separately as VIX_LEVEL for later forecasting controls, but it is not part of the six-dimensional Financial DNA vector.

The factors are intentionally expressed in forms that reflect their economic meaning. For example, the RATE factor measures the daily change in the US 10-year Treasury yield rather than the return of a bond ETF, while the CREDIT factor compares high-yield corporate bonds with government-bond exposure. The resulting factor vector therefore captures changes in market conditions rather than simply combining six asset returns.

For each asset, the sensitivity to these factors is estimated through a rolling regression. The resulting six coefficients form the asset's time-varying Financial DNA vector, which becomes the central input for the structural-drift measures developed in the following chapters.

2.3 Sample Period and Data Frequency

The empirical sample covers the period from 4 January 2010 to 30 September 2026 and is constructed at a daily frequency. After aligning the asset and factor series on common trading dates, the resulting analysis dataset contains approximately 4,146 daily observations.

Daily data are used because the objective is to capture gradual changes in factor sensitivities while retaining enough observations for rolling estimation. A 252-trading-day window is used for the rolling factor regressions, corresponding approximately to one year of market data. A minimum of 200 valid observations is required before a rolling estimate is retained.

The full sample is subsequently divided into distinct analytical periods. Earlier observations are used to establish historical reference structures, while later periods are reserved for model development and genuine out-of-sample evaluation. This separation is important because it prevents the forecasting analysis from being assessed on the same observations used to estimate the model.

Period	Dates	Purpose
Full sample	2010-01-04 to 2026-09-30	Construction of returns, factors and rolling exposures
Fixed reference period	2011-01-01 to 2014-12-31	Calibration of the original fixed-distance MIDI
Development period	2015-01-01 to 2021-12-31	Model development, percentile reference and training
Holdout period	2022-01-01 to 2026-09-30	Genuine out-of-sample evaluation

Note: The Dynamic MIDI requires additional rolling history and therefore becomes available later than the raw return dataset. Training and reference periods are always defined using information available prior to the relevant evaluation period.

This chronological structure is maintained throughout the project. In particular, all reported out-of-sample results use coefficients estimated only from earlier observations, and the expanding-window validation further updates the training sample sequentially without using future information.

2.4 Return and Factor Construction

All price-based series are transformed into daily log returns before entering the factor model. Log returns provide an additive representation through time and are convenient for rolling regression analysis. Factors expressed naturally as changes in levels, such as interest rates and the VIX, are treated separately.

The transformations are designed to preserve the economic interpretation of each factor while ensuring that the same factor definitions can be applied consistently across the six assets.

2.4.1 Equity and Asset Returns

Daily returns for QQQ, XLF, XLE, FEZ, TLT and GLD are calculated from adjusted closing prices using continuously compounded returns. The same transformation is applied to the price-based market factors VT, UUP and crude oil futures.

$$r_t = \ln\left(\frac{P_t}{P_{t-1}}\right)$$

where P_t denotes the adjusted closing price at date t

The broad market factor, MKT, is defined as the daily log return of VT, a global equity-market proxy. USD is defined from the daily log return of UUP, while OIL is based on the daily log return of front-month WTI crude oil futures.

2.4.2 Interest Rate Factor

The interest-rate factor is constructed from the US 10-year Treasury constant-maturity yield. Because the yield series is expressed in percentage points, its daily change is multiplied by 100 so that the resulting factor is measured in basis points.

$$RATE_t = 100(DGS10_t - DGS10_{t-1})$$

A positive RATE observation therefore represents an increase in long-term yields, while a negative observation represents a decline. For example, a move from 4.20% to 4.25% corresponds to a RATE value of +5 basis points.

2.4.3 Credit Factor

Credit conditions are represented through the relative performance of high-yield corporate bonds and intermediate US Treasury bonds. Rather than using a quoted credit spread series directly, the project constructs a market-based credit factor from the return differential between HYG and IEF.

$$CREDIT_t = r_{HYG,t} - r_{IEF,t}$$

A positive CREDIT observation indicates that high-yield bonds outperformed government bonds during the day, whereas a negative value indicates relative underperformance of credit-sensitive securities. The factor therefore captures changes in credit-market conditions through traded asset prices.

2.4.4 Volatility Factor

Market uncertainty is represented using the CBOE Volatility Index. Within the Financial DNA regression, the volatility factor is defined as the daily change in the VIX rather than its level.

$$VIX_t = VIXLEVEL_t - VIXLEVEL_{t-1}$$

This specification captures shocks to implied market volatility. The absolute VIX level is retained separately as VIX_LEVEL and is later used as a control variable in forecasting regressions. It does not form part of the six-dimensional Financial DNA vector.

Factor	Mathematical Construction	Unit
MKT	$r_{VT,t}$	Log return
RATE	$100(DGS10_t - DGS10_{t-1})$	Basis points
CREDIT	$r_{HYG,t} - r_{IEF,t}$	Return differential
USD	$r_{UUP,t}$	Log return
OIL	$r_{CL,t}$	Log return
VIX	$VIXLEVEL_t - VIXLEVEL_{t-1}$	Index-point change

Note: VIX_LEVEL is retained as a separate forecasting control and is not included in the Financial DNA vector.

After these transformations, all asset returns and factors are aligned on common trading dates to form the daily analysis dataset. This dataset constitutes the input for the rolling multi-factor regressions developed in the next chapter.

3. Modelling Financial DNA

The central idea of Market DNA is that the economic identity of a financial asset can be represented by its sensitivities to a common set of market factors. Rather than assuming these sensitivities to be constant, the framework allows them to evolve through time.

Each asset is therefore characterised by a six-dimensional vector of rolling factor exposures. Changes in this vector represent changes in the way the asset interacts with broader financial conditions. The purpose of this chapter is to estimate these evolving exposures and transform them into quantitative measures of structural identity drift.

3.1 Rolling Factor Exposures

For each of the six assets, daily returns are regressed on the six market factors defined in Chapter 2. The model is estimated repeatedly through a rolling window, allowing the factor coefficients to change through time rather than imposing a single relationship over the full sample.

For asset i at date t, the underlying factor model is:

$$r_{i,t} = \alpha_{i,t} + \beta_{MKT,i,t}MKT_t + \beta_{RATE,i,t}RATE_t + \beta_{CREDIT,i,t}CREDIT_t + \beta_{USD,i,t}USD_t + \beta_{OIL,i,t}OIL_t + \beta_{VIX,i,t}VIX_t + \varepsilon_{i,t}$$

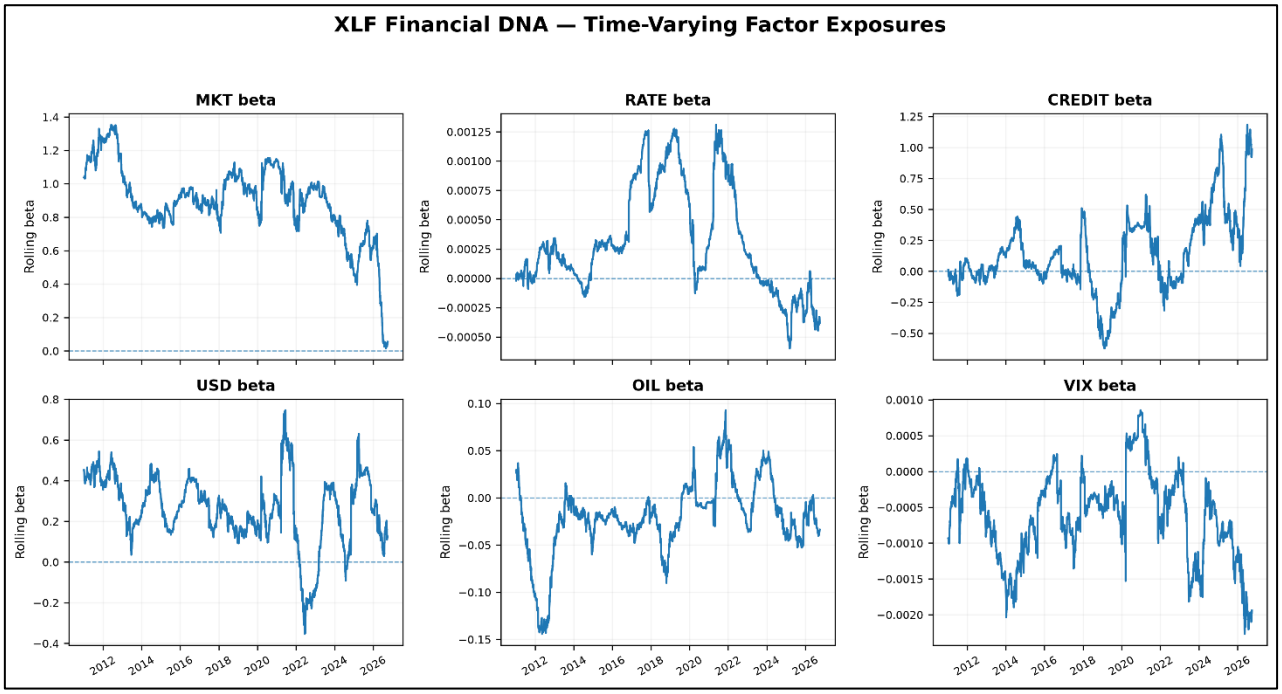
where $r_{i,t}$ is the daily return of asset i, $\alpha_{i,t}$ is the rolling intercept, and each β measures the asset's estimated sensitivity to one of the six common factors. The residual term $\varepsilon_{i,t}$ captures the part of the asset return not explained by the factor model.

The regressions are estimated using a 252-trading-day rolling window, approximately corresponding to one year of observations. At least 200 valid daily observations are required for an estimate to be retained. As the window moves forward one trading day at a time, a new set of factor coefficients is obtained.

This procedure produces six time series of rolling betas for every asset. A positive beta indicates that the asset has tended to move in the same direction as the corresponding factor within the estimation window, while a negative beta indicates an inverse relationship. The magnitude of the coefficient reflects the strength of that sensitivity, although coefficients are not directly comparable across factors when the factor units differ.

The rolling betas should be interpreted as **conditional statistical exposures rather than causal effects**. A change in a beta indicates that the empirical relationship between an asset and a factor has changed within the rolling window; it does not by itself establish that the factor caused the asset's return.

Figure 1. Example of Time-Varying Financial DNA: Rolling Factor Exposures of XLF



Note: Each coefficient is estimated using a 252-trading-day rolling OLS regression. The figure illustrates that factor sensitivities are not constant through time, providing the empirical motivation for treating Financial DNA as a dynamic object.

The rolling estimates show substantial variation through time. In particular, exposures to interest rates, credit conditions, the US dollar and market volatility can change both in magnitude and, in some cases, in sign. This time variation provides the empirical foundation for the Market DNA framework: if factor sensitivities were stable, there would be little reason to construct a structural-drift measure.

However, observing six changing coefficients separately does not provide a single measure of how much an asset's overall identity has changed. The next sections therefore treat the six betas jointly as a multivariate Financial DNA vector.

3.2 Financial DNA as a Time-Varying Exposure Vector

The six rolling factor coefficients can be interpreted jointly rather than as independent statistics. At each date t, asset i is represented by a six-dimensional exposure vector containing its contemporaneous sensitivities to market, interest-rate, credit, US-dollar, oil and volatility factors.

This vector constitutes the asset's **Financial DNA** at that point in time.

$$\boldsymbol{\beta}_{i,t} = \left(\beta_{MKT,i,t}, \beta_{RATE,i,t}, \beta_{CREDIT,i,t}, \beta_{USD,i,t}, \beta_{OIL,i,t}, \beta_{VIX,i,t}\right)^{T}$$

where $\boldsymbol{\beta}_{i,t}$ represents the Financial DNA of asset i at date t. Each component describes one dimension of the asset's relationship with the broader financial environment.

Financial identity drift occurs when this vector changes materially through time. A small movement in one coefficient may represent normal estimation noise, whereas simultaneous or persistent changes across several factor sensitivities may indicate a more fundamental reconfiguration of the asset's economic behaviour.

Importantly, measuring the distance between two Financial DNA vectors is not as simple as comparing the coefficients one by one. The six factor exposures may have different levels of historical variability and may also move together. For example, changes in market and credit sensitivities may be correlated, while another exposure may historically remain much more stable.

A useful structural-drift measure should therefore account for both the historical **variance of each beta and the covariance between betas.** This requirement motivates the use of a multivariate Mahalanobis distance in the next section.

> **Why not use a simple Euclidean distance?**
> A one-unit movement is not equally unusual for every beta. In addition, factor exposures may co-move. Mahalanobis distance adjusts the measured drift for both the historical variability of each exposure and the covariance structure between exposures.

3.3 Fixed Market Identity Drift

Once Financial DNA has been represented as a multivariate exposure vector, the next step is to quantify how far the current structure has moved from a historical reference. The first version of the Market DNA framework uses **a fixed reference period from 2011 to 2014.**

For each asset, the rolling Financial DNA vectors observed during this reference period are used to estimate a historical mean vector and covariance matrix. These two objects define the asset's baseline factor structure: the mean describes its typical **historical** exposures, while the covariance matrix describes how those exposures normally vary and co-move.

Let μ_i denote the mean Financial DNA vector of asset i during the reference period, and let Σ_i denote the corresponding covariance matrix.

The distance between the current Financial DNA vector and its historical reference is measured using the Mahalanobis distance. Unlike a simple Euclidean distance, this measure standardises deviations according to the historical variability of each exposure and accounts for correlations between factor sensitivities.

$$\boldsymbol{IDS_{i,t}} = \sqrt{\left((\boldsymbol{\beta_{i,t}} - \boldsymbol{\mu_i})^{\boldsymbol{T}} \boldsymbol{\Sigma_i^{-1}} (\boldsymbol{\beta_{i,t}} - \boldsymbol{\mu_i})\right)}$$

where $IDS_{i,t}$ is the Identity Drift Score of asset i at date t. A low score indicates that the asset's current factor structure remains close to its historical reference, while a high score indicates that the combination of factor sensitivities has moved unusually far from that baseline.

This distinction is important because the six beta coefficients do not have identical statistical properties. A change of a given magnitude may be highly unusual for one exposure but entirely normal for another. Furthermore, several betas may move together during normal market conditions. Mahalanobis distance adjusts for both effects before determining whether the current Financial DNA is unusual.

The six asset-level Identity Drift Scores are aggregated into a single cross-asset indicator called the **Market Identity Drift Index (MIDI)**. At each date, MIDI is calculated as the equally weighted average of the six individual drift scores.

$$\boldsymbol{MIDI_t} = \frac{\boldsymbol{1}}{\boldsymbol{N}} \sum_{\boldsymbol{i=1}}^{\boldsymbol{N}} \boldsymbol{IDS_{i,t}} \qquad \boldsymbol{N = 6}$$

The use of an equal-weighted average ensures that no asset is mechanically given greater importance than another. MIDI therefore summarises the average multivariate distance of the six assets from their respective historical identities.

To facilitate interpretation, the distribution of MIDI during the fixed calibration period is used to define a historical benchmark. The 95th percentile of the calibration distribution is approximately 2.93. Values

above this level therefore represent unusually large structural deviations relative to the 2011–2014 reference environment.

Empirically, the fixed MIDI rises substantially over the later part of the sample. By September 2026, the index is far above its original calibration range. This result initially appears to indicate very large market identity drift.

However, the interpretation requires caution. Because the reference structure is permanently anchored to 2011–2014, the indicator increasingly measures the distance between the current market environment and an old historical regime. A persistently high MIDI may therefore reflect long-term structural evolution rather than an acute or temporary market dislocation.

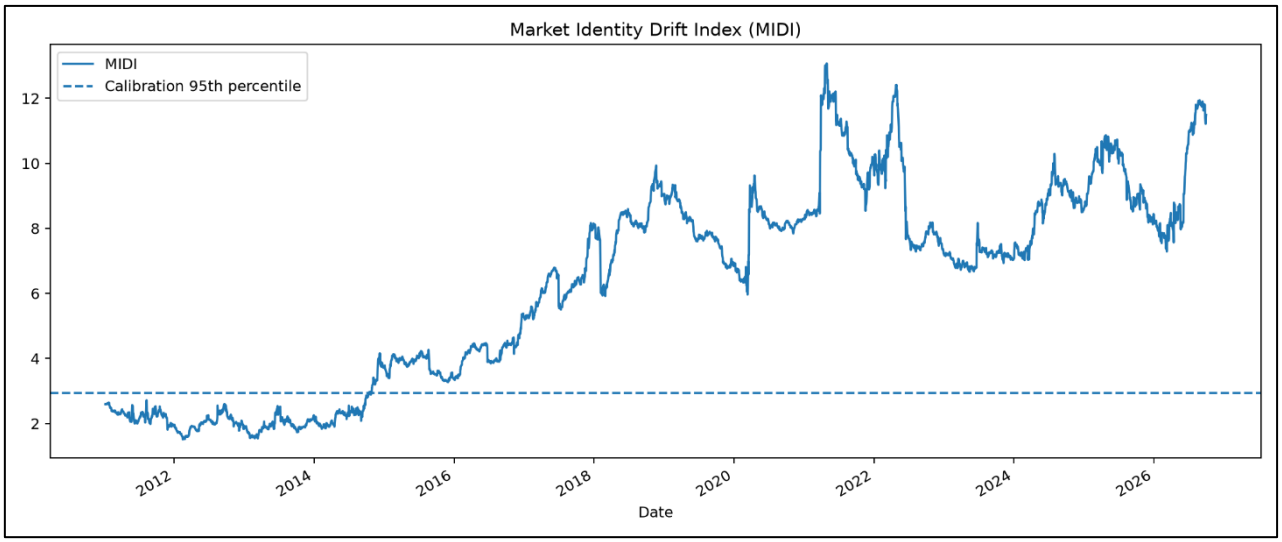
Interpretation of Fixed MIDI

A high fixed MIDI does not necessarily indicate market stress. It indicates that current factor exposures are far from the 2011–2014 reference structure. Persistent structural change can therefore keep the index elevated even in the absence of an immediate crisis.

The fixed-reference approach is useful because it provides a transparent measure of long-term structural transformation. However, it becomes less suitable when the objective is to identify whether current exposures are unusual relative to the recent market environment.

This limitation motivates the Dynamic Market Identity Drift framework developed in the next section. Instead of comparing current Financial DNA with a permanently fixed historical baseline, the dynamic approach updates the reference structure through time while maintaining a deliberate gap between the reference window and the current observation.

Figure 2. Fixed Market Identity Drift Index Relative to the 2011–2014 Reference



Note: MIDI is the equally weighted average of the six asset-level Mahalanobis Identity Drift Scores. The dashed line represents the 95th percentile of the 2011–2014 calibration distribution.

Figure 2 shows that the fixed-reference MIDI remains relatively low during the calibration period but rises substantially over the later sample. By the end of the sample, the index stands far above its historical 95th-percentile threshold. This confirms that the factor structure of the six-asset universe has moved materially away from the 2011–2014 reference environment.

However, this result must be interpreted carefully. Because the reference period is permanently fixed, the indicator increasingly measures the distance between the current market structure and an older historical

regime. A persistently elevated fixed MIDI may therefore capture long-term structural change rather than an acute temporary disruption.

3.4 Dynamic Market Identity Drift

The fixed-reference MIDI provides a useful measure of long-term structural change, but its interpretation becomes increasingly dependent on the choice of the original 2011–2014 calibration period. To identify whether current Financial DNA is unusual relative to a more recent market environment, the framework is therefore extended to a dynamic reference structure.

The Dynamic Market Identity Drift framework updates the historical benchmark through time. Each current Financial DNA vector is compared with a rolling distribution of past factor exposures rather than with a permanently fixed baseline. This produces an asset-level Dynamic Identity Drift Score (DIDS) and an aggregate Dynamic Market Identity Drift Index (DMIDI).

3.4.1 Rolling Reference Window

For each date t, the dynamic reference distribution is constructed using **504 historical rolling Financial DNA observations**, corresponding approximately to two trading years. Importantly, the reference window does not extend directly to the current observation. A **252-trading-day gap** is imposed between the end of the reference period and date t.

The gap reduces the mechanical similarity between the current rolling regression and the observations used to define its historical benchmark. This is particularly relevant because each Financial DNA vector is itself estimated using a 252-day rolling window.

The dynamic benchmark therefore represents a recent but sufficiently separated historical structure against which current factor exposures can be evaluated.

3.4.2 Mahalanobis Distance

As in the fixed-reference framework, structural distance is measured using the Mahalanobis metric. However, both the reference mean vector and covariance matrix now vary through time.

$$\boldsymbol{DIDS_{i,t}} = \sqrt{\left(\left(\boldsymbol{\beta_{i,t}} - \boldsymbol{\mu_{i,t}}\right)^{T} \boldsymbol{\Sigma_{i,t}^{-1}} \left(\boldsymbol{\beta_{i,t}} - \boldsymbol{\mu_{i,t}}\right)\right)}$$

where $\mu_{i,t}$ and $\sum_{i,t}$ are respectively the mean vector and covariance matrix estimated from the rolling historical reference window for asset i. Because these reference parameters evolve through time, the same absolute change in factor exposures may receive a different drift score depending on the prevailing historical structure.

3.4.3 Asset-Level Dynamic Identity Drift Scores

The resulting DIDS measures how unusual the current Financial DNA of each asset is relative to its own recent historical distribution. A high value indicates that the joint configuration of factor sensitivities lies far from the rolling reference structure.

The measure remains multivariate: an elevated DIDS can result from a very large change in one exposure, from simultaneous changes across several exposures, or from a combination of deviations that is historically unusual given the covariance structure of the betas.

This asset-level decomposition is important because two market environments can display a similar aggregate level of structural drift while being driven by very different assets.

3.4.4 Aggregate DMIDI

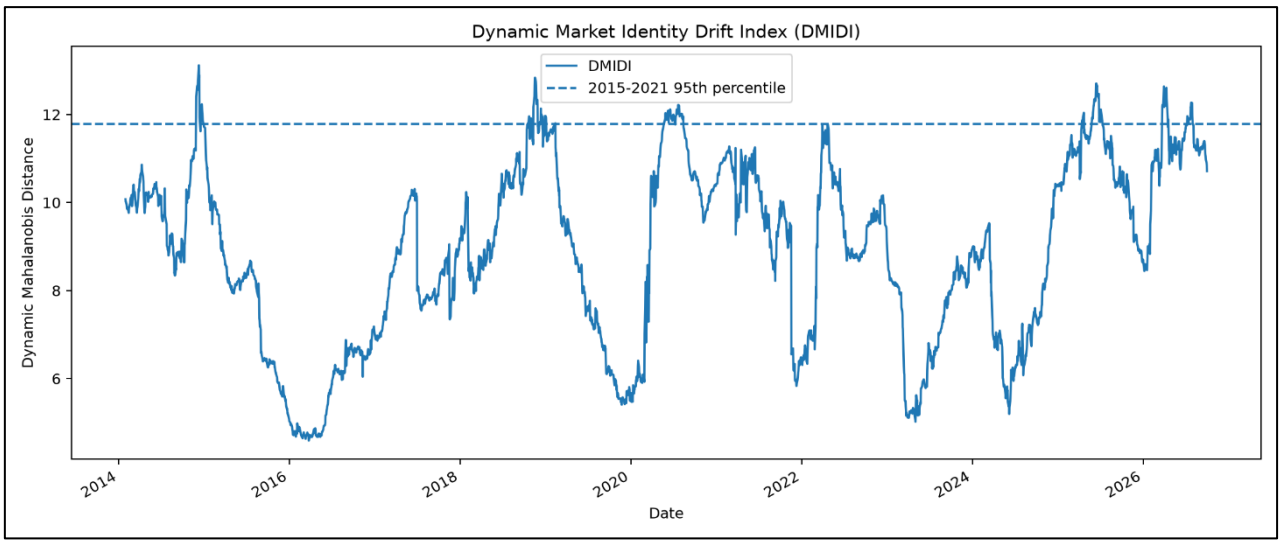
The six Dynamic Identity Drift Scores are aggregated into the Dynamic Market Identity Drift Index (DMIDI). As with the fixed MIDI, each asset receives an equal weight.

$$DMIDI_t = \frac{1}{N}\sum_{i=1}^{N} DIDS_{i,t} \qquad N = 6$$

DMIDI measures the average degree of structural displacement across the six assets relative to their respective rolling reference distributions. Unlike fixed MIDI, it is designed to adapt gradually as the underlying financial environment evolves.

This distinction changes the interpretation of the indicator. A high fixed MIDI means that markets are far from the original 2011–2014 structure. **A high DMIDI instead indicates that current factor exposures are unusual relative to a more recent historical benchmark.**

Figure 3. Dynamic Market Identity Drift Index



Note: DMIDI is the equally weighted average of the six asset-level Dynamic Identity Drift Scores. Each DIDS compares the current Financial DNA vector with a 504-observation rolling historical reference separated from the current observation by a 252-trading-day gap.

Figure 3 shows that dynamic structural drift varies considerably through time rather than following the persistent upward pattern observed in the fixed-reference MIDI. This confirms that part of the increase in fixed MIDI reflects long-term distance from the original calibration regime rather than continuously increasing short-term structural instability.

The dynamic specification therefore provides a more suitable measure for identifying episodes in which current Financial DNA becomes unusual relative to recent history.

At the end of September 2026, DMIDI is approximately **10.71**, indicating elevated structural drift relative to the rolling historical benchmark. However, the aggregate level alone does not reveal whether this drift is broadly distributed across the market or concentrated in a small number of assets.

Measuring the magnitude of structural drift is therefore only the first step. The next chapter decomposes the aggregate signal across assets and introduces Drift Breadth to distinguish concentrated structural changes from broad market-wide transformations.

4. From Drift to Market Structure

DMIDI summarises the average magnitude of structural identity drift across the six assets, but an aggregate value alone does not reveal how that drift is distributed. A high DMIDI may result from an extreme change in a single asset or from moderate changes occurring simultaneously across several assets.

Distinguishing between these configurations is important because they describe different forms of market transformation. This chapter therefore decomposes aggregate structural drift across assets and introduces a measure of Drift Breadth to distinguish concentrated from broad structural change.

4.1 Asset-Level Drift Attribution

Because DMIDI is constructed from the six asset-level Dynamic Identity Drift Scores, the aggregate signal can be decomposed directly into individual asset contributions. At each date, the relative contribution of asset i is defined as its DIDS divided by the sum of the six DIDS values.

$$w_{i,t} = \frac{DIDS_{i,t}}{\sum_{j=1}^{N} DIDS_{j,t}} \qquad N = 6$$

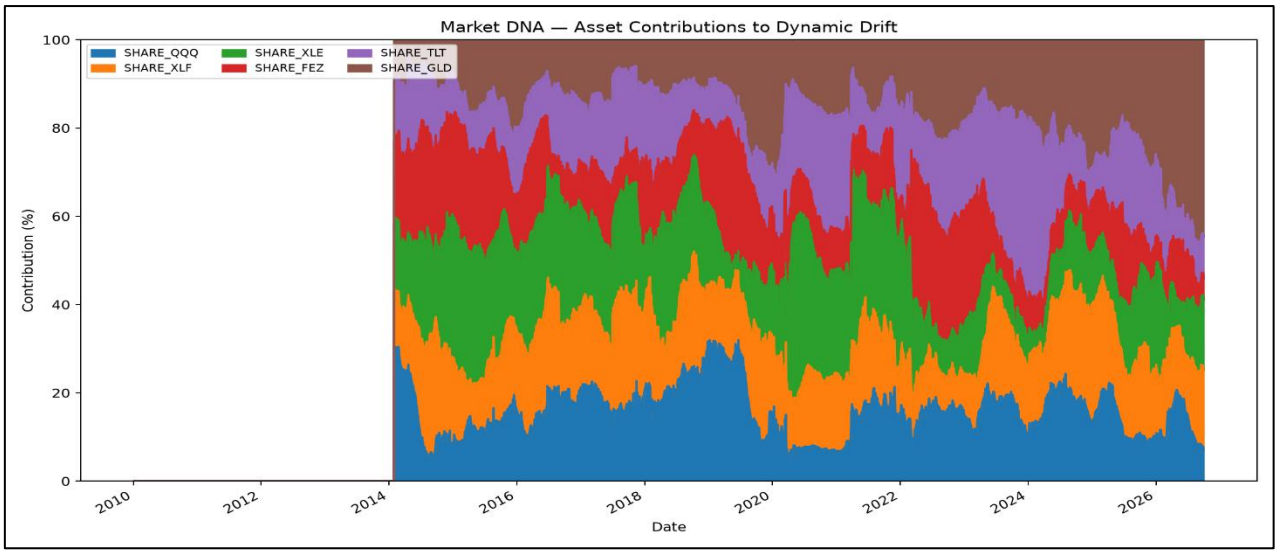
where $w_{i,t}$ is the share of total structural drift attributable to asset i at date t. By construction, the contribution shares sum to one across the six assets.

$$\sum_{i=1}^{N} w_{i,t} = 1$$

These contribution shares provide an intuitive attribution layer for the Market DNA framework. When one asset accounts for a large fraction of total DIDS, the aggregate structural signal is concentrated. Conversely, when contributions are relatively balanced, structural change is more broadly distributed across the cross-asset universe.

The decomposition is descriptive rather than causal. A large contribution does not imply that the asset caused the broader market transition; it indicates that its Financial DNA is unusually distant from its recent historical structure relative to the other assets.

Figure 4. Asset Contributions to Dynamic Market Identity Drift



Note: Contributions are calculated from each asset's share of the sum of the six Dynamic Identity Drift Scores at each date. Higher shares indicate that a larger proportion of aggregate structural drift is associated with that asset.

Figure 4 illustrates that the composition of structural drift changes materially through time. Different episodes are dominated by different assets, while other periods exhibit a more balanced distribution of contributions. This reinforces the idea that similar levels of DMIDI can correspond to economically different market configurations.

Asset contributions identify **who is drifting**, but they do not yet provide a single measure of whether structural change is concentrated or widespread. This motivates the introduction of Drift Breadth in the next section

4.2 Drift Breadth

DMIDI measures the average magnitude of structural identity drift across the six assets, but it does not indicate whether this drift is concentrated in a small number of assets or broadly distributed across the cross-asset universe.

To capture this distinction, the Market DNA framework introduces **Drift Breadth**, a normalized entropy measure based on the asset-level contribution shares defined in the previous section. Drift Breadth is designed to separate concentrated structural changes from broad market-wide transformations.

4.2.1 Contribution Shares

For each asset, the contribution share represents its proportion of total dynamic identity drift at date t. The six contribution shares therefore describe how aggregate structural drift is distributed across the asset universe.

$$\boldsymbol{w_{i,t} = \frac{DIDS_{i,t}}{\sum_{j=1}^{N} DIDS_{j,t}}}$$

where $w_{i,t}$ is the contribution share of asset i, with $N = 6$. By construction, the shares are non-negative and sum to one.

$$\boldsymbol{\sum_{i=1}^{N} w_{i,t} = 1}$$

A highly concentrated structural episode is characterised by one or two large contribution shares, whereas a broad structural episode produces a more balanced distribution across the six assets.

4.2.2 Entropy-Based Breadth Measure

The distribution of the contribution shares is summarised using normalized Shannon entropy. Entropy is well suited to this purpose because it increases when contributions become more evenly distributed and decreases when structural drift is dominated by a small number of assets.

$$\boldsymbol{DRIFT_{BREADTH_t} = -\frac{\sum_{i=1}^{N} w_{i,t} \ln(w_{i,t})}{\ln(N)}}$$

Dividing by $\ln(N)$ normalizes the measure to the interval between zero and one.

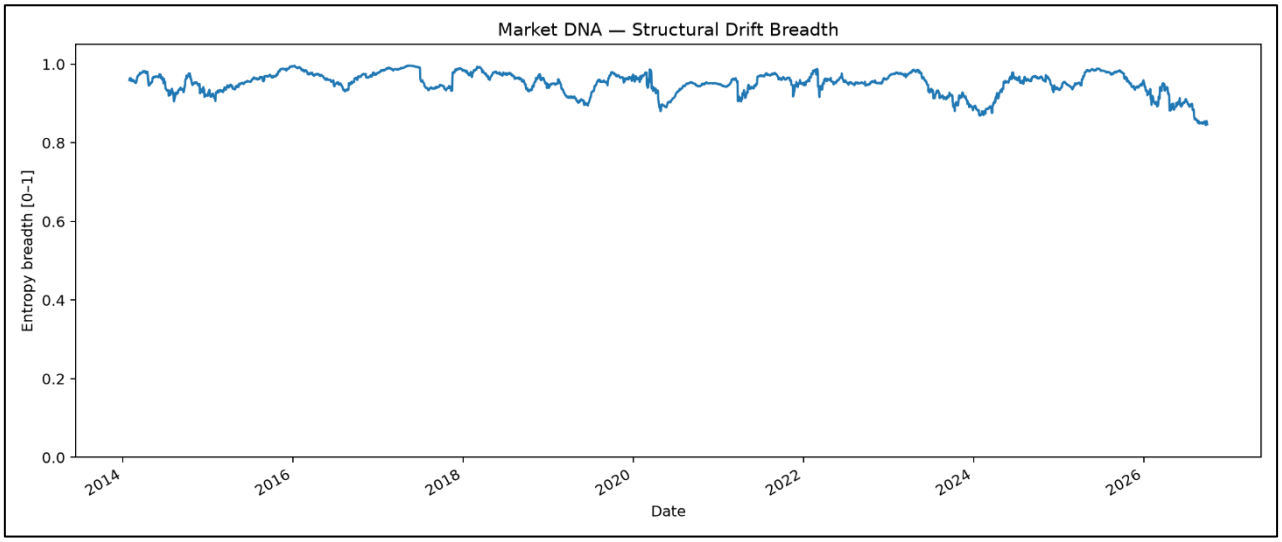
$$\boldsymbol{0 \leq DRIFT_{BREADTH_t} \leq 1}$$

A Drift Breadth value close to **zero** indicates that structural change is highly concentrated in a small number of assets. In the extreme case where one asset accounts for virtually all observed drift, entropy approaches zero.

A value close to **one** indicates that structural drift is distributed relatively evenly across the asset universe. In this case, several assets are simultaneously experiencing unusual changes in their Financial DNA.

Drift Breadth should therefore be interpreted jointly with DMIDI. A high DMIDI combined with low breadth describes a large but concentrated structural change, while a high DMIDI combined with high breadth indicates a broad cross-asset transformation.

Figure 5. Evolution of Cross-Asset Drift Breadth



Note: Drift Breadth is calculated from the normalized entropy of the six asset-level Dynamic Identity Drift contribution shares. Values closer to one indicate a broader distribution of structural drift across assets, while lower values indicate greater concentration.

Figure 5 shows that the distribution of structural drift varies through time. Some episodes are characterised by relatively balanced contributions across assets, while others are dominated by a narrower subset of the cross-asset universe. This information cannot be recovered from DMIDI alone.

The combination of drift magnitude and drift breadth therefore provides a richer description of market structure than either measure considered separately.

The next step is to examine whether asset-level structural drift contains a common component across markets or remains predominantly asset-specific.

4.3 Common Structural Drift

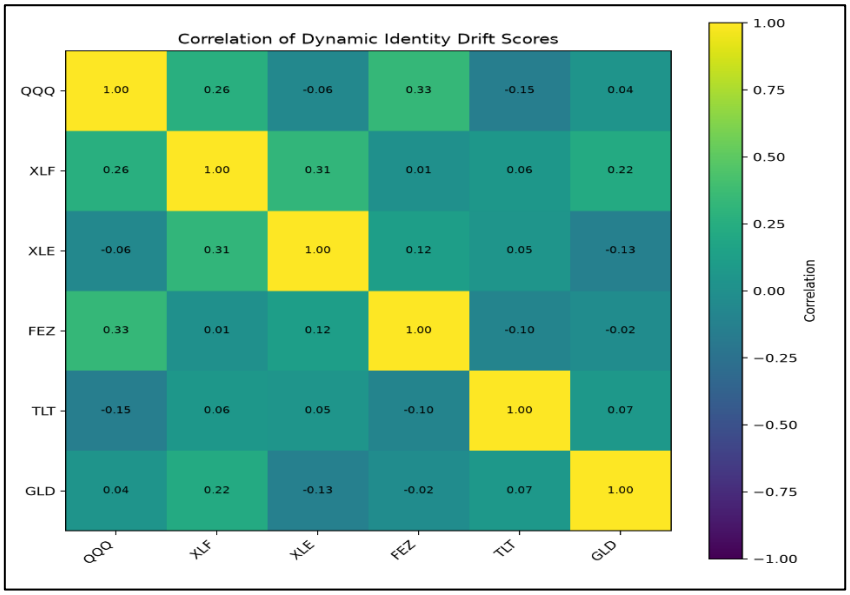
Drift Breadth describes how structural change is distributed across assets at a given date, but it does not indicate whether the underlying asset-level drift measures systematically move together through time. To investigate this question, the analysis combines pairwise correlations of the Dynamic Identity Drift Scores with principal component analysis.

These two approaches provide complementary information. Correlations measure the strength of bilateral co-movement between asset-level drift scores, while principal component analysis evaluates whether a common latent component can explain a substantial share of their joint variation.

4.3.1 Correlation of Asset-Level Drift Scores

The first step is to examine the pairwise correlation structure of the six DIDS series. If structural identity drift were predominantly driven by a single market-wide force, consistently high positive correlations would be expected across assets. Conversely, weak or moderate correlations would suggest that structural change often remains asset-specific or occurs through different transmission channels.

Figure 6. Correlation Matrix of Asset-Level Dynamic Identity Drift Scores



Note: The matrix reports pairwise correlations between the six asset-level Dynamic Identity Drift Score series. Higher positive values indicate that structural drift tends to increase simultaneously across the corresponding assets.

Figure 6 shows that the correlations between asset-level drift scores are generally weak to moderate rather than uniformly high. For example, the correlation between QQQ and FEZ is approximately **0.33**, while XLF and XLE exhibit a correlation of about **0.31**. The correlation between QQQ and XLF is approximately **0.26**.

These relationships indicate that some common structural movement exists, particularly across equity-related assets, but the evidence does not support the idea that identity drift is driven by a single synchronized market process. Different assets can experience substantial changes in their factor structure at different times.

4.3.2 Principal Component Analysis

To assess the presence of a broader common component more systematically, principal component analysis (PCA) is applied to the six standardized DIDS series. Standardization is performed before the PCA so that differences in the scale and historical dispersion of the individual drift scores do not mechanically determine the results.

$$z_{i,t} = \frac{DIDS_{i,t} - bar(DIDS_i)}{s_i}$$

The first principal component explains approximately 25.38% of the total variation in the standardized asset-level drift scores. This proportion is meaningful, but it is far from dominant. The remaining variation is therefore distributed across several additional dimensions. The estimated first-component loadings are shown below.

Table 5. First Principal Component Loadings

Asset	PC1 Loading
QQQ	0.729
XLF	0.659
XLE	0.378
FEZ	0.589
TLT	-0.186
GLD	0.181

Note: PCA is performed on standardized DIDS series. The overall sign of a principal component is arbitrary; interpretation should therefore focus on the relative signs and magnitudes of the loadings.

QQQ, XLF and FEZ have the largest positive loadings on the first component, suggesting that part of their structural drift tends to share a common cross-equity dimension. XLE also loads positively, although less strongly, while TLT exhibits a small loading of the opposite sign and GLD contributes only modestly to this first component.

More importantly, the first component explains only around one quarter of total DIDS variation. Structural identity drift is therefore **multidimensional**: a common market component exists, but a large majority of the variation cannot be reduced to a single systematic factor.

Taken together, Drift Contributions, Drift Breadth, the correlation structure and PCA show that aggregate Market DNA drift can arise through materially different configurations. The same level of DMIDI may reflect a concentrated shift in one asset, a broad but heterogeneous transformation across several assets, or a partially synchronized cross-market movement.

This distinction becomes important when assessing whether structural drift contains information about subsequent market risk. The next chapter therefore moves from structural description to formal predictive testing.

5. Does Market DNA Predict Market Risk?

The previous chapters show that Financial DNA changes through time and that structural drift can be measured, decomposed and characterised across assets. A natural next question is whether these structural changes contain information about subsequent market conditions.

The initial hypothesis of the project was that unusually large identity drift might precede periods of elevated market risk. This hypothesis is tested explicitly rather than assumed. The analysis focuses primarily on two forward-looking outcomes: future realised volatility and future cross-asset correlation.

Particular attention is given to out-of-sample evaluation. A statistically interesting relationship within the historical sample does not necessarily imply useful predictive information. The forecasting exercises therefore compare Market DNA indicators with simpler benchmark models and evaluate whether any improvement survives when only information available at the time of prediction is used.

5.1 Forecast Targets

Two dimensions of future market risk are examined. The first is **future realised volatility**, which measures the magnitude of subsequent market fluctuations. The second is **future cross-asset correlation**, which captures the degree to which the assets subsequently move together.

These two targets represent different dimensions of risk. Volatility describes the intensity of price movements, while correlation describes the degree of diversification available across assets. Structural identity drift could, in principle, affect either dimension without necessarily affecting the other.

Unless otherwise specified, the primary forecasting horizon is **20 trading days**, approximately one market month. Predictive regressions are estimated using lagged information so that future outcomes do not enter the explanatory variables.

5.2 Future Volatility

The first forecasting exercise investigates whether Market DNA improves the prediction of future realised volatility. This represents a deliberately demanding test because volatility is already strongly related to established indicators such as the VIX and recent realised volatility.

The objective is therefore not simply to determine whether DMIDI is correlated with future volatility, but whether it provides **incremental predictive information** beyond these conventional controls.

5.2.1 In-Sample Evidence

Initial descriptive analysis suggests a weak positive relationship between Dynamic Market Identity Drift and subsequent volatility. Rank correlations between DMIDI and future realised volatility are positive across several horizons, with the strongest values around the 10- to 20-day range.

However, these correlations are modest and do not control for other variables that are already known to contain information about future volatility. More formal regression analysis is therefore required before interpreting this relationship as predictive.

The forecasting specification controls for the contemporaneous VIX level and recent realised volatility. A generic 20-day forecasting regression can be written as:

$$\boldsymbol{RV_{t+1:t+20} = \alpha + \beta_1 DMIDI_t + \beta_2 VIXLEVEL_t + \beta_3 PASTRV_t + \varepsilon_t}$$

where $RV_{t+1,t+20}$ denotes realised volatility over the following 20 trading days, $VIXLEVEL_t$ is the current VIX level and $PASTRV_t$ represents recent realised volatility.

Once these controls are introduced, the predictive contribution of DMIDI becomes weak. This already suggests that much of the apparent relationship between structural drift and future volatility overlaps with information contained in conventional market-risk indicators.

A nonlinear specification was also tested by including a squared DMIDI term. Neither the linear nor the quadratic DMIDI coefficient is statistically significant once the VIX and past realised volatility are included.

In the quadratic specification, the linear DMIDI coefficient has a p-value of approximately **0.32**, while the squared term has a p-value of approximately **0.49**. By contrast, the VIX remains strongly significant.

The apparent nonlinear pattern visible in descriptive volatility plots should therefore not be interpreted as robust predictive evidence. This distinction between descriptive shape and statistically validated forecasting information is maintained throughout the project.

5.2.2 Out-of-Sample Volatility Forecasting

To determine whether the apparent relationship between structural drift and future volatility has genuine predictive value, the models are evaluated out of sample. The estimation period covers **2015–2021**, while the period from **2022 to September 2026** is reserved as a holdout sample.

Model coefficients are therefore estimated exclusively using the training period and are then applied to observations that were not used during estimation. This provides a more demanding test than in-sample statistical significance because the model must perform on genuinely unseen data.

Forecast accuracy is evaluated primarily using the Root Mean Squared Error (RMSE). Lower RMSE values indicate more accurate volatility forecasts.

$$RMSE = \sqrt{\left(\frac{1}{T}\sum_{t=1}^{T}(y_t - hat(y)_t)^2\right)}$$

where y_t is the realised future volatility and $\hat{y}_t$ is the corresponding model forecast.

Table 6. Out-of-Sample Volatility Forecast Performance

Model	RMSE	OOS R^2 vs. VIX	Interpretation
VIX benchmark	0.0502	—	Best simple benchmark
VIX + Past RV	0.0503	—	No material improvement
VIX + Past RV + Fixed MIDI	0.0534	-0.132	Worse than VIX benchmark
VIX + Past RV + DMIDI	0.0551	-0.206	Worse than VIX benchmark

Note: The holdout period runs from 2022 to September 2026. Negative out-of-sample R^2 values indicate that the corresponding Market DNA specification produces larger forecast errors than the VIX benchmark.

The results are unambiguous. The simple VIX benchmark achieves an RMSE of approximately **0.0502**. Adding past realised volatility does not materially improve forecast accuracy, while introducing either fixed MIDI or dynamic DMIDI increases the forecast error.

The fixed-MIDI specification produces an RMSE of approximately **0.0534**, corresponding to an out-of-sample R^2 of about **−0.13** relative to the VIX benchmark. The dynamic specification performs even less favourably, with an RMSE of approximately **0.0551** and an out-of-sample R^2 of about **−0.21**.

These results reject the idea that Market DNA provides a superior standalone signal for forecasting future realised volatility. Structural identity drift and volatility measure different aspects of market behaviour, and large changes in factor relationships do not necessarily imply larger subsequent price fluctuations.

Structural identity drift and volatility measure different aspects of market behaviour.

This negative result is informative rather than problematic. The objective of the project is not to preserve the initial hypothesis at all costs, but to determine empirically where the Market DNA framework adds information and where it does not. The volatility tests indicate that its main value is unlikely to lie in replacing conventional volatility indicators such as the VIX.

$$R^2_{OOS} = 1 - \frac{MSE_{model}}{MSE_{benchmark}}$$

A positive value indicates an improvement over the benchmark, while a negative value indicates weaker out-of-sample performance.

The failure to improve volatility forecasts leads to an important refinement of the research question. Market DNA may not predict how much markets will move, but structural changes in asset relationships may still contain information about **how markets move together**. The analysis therefore turns next to future cross-asset correlation.

5.3 Future Cross-Asset Correlation

The second forecasting target is future cross-asset correlation. Whereas volatility measures the magnitude of subsequent price movements, correlation captures the extent to which different assets subsequently move together. This dimension is particularly relevant for diversification because portfolio risk can increase even without a large rise in individual asset volatility when correlations become more positive.

The analysis therefore investigates whether structural identity drift contains information about future co-movement across the six-asset universe.

Future correlation is measured over a **20-trading-day horizon**, using the subsequent return behaviour of the six assets. The predictive models include controls for recent cross-asset correlation, the current VIX level and past realised volatility.

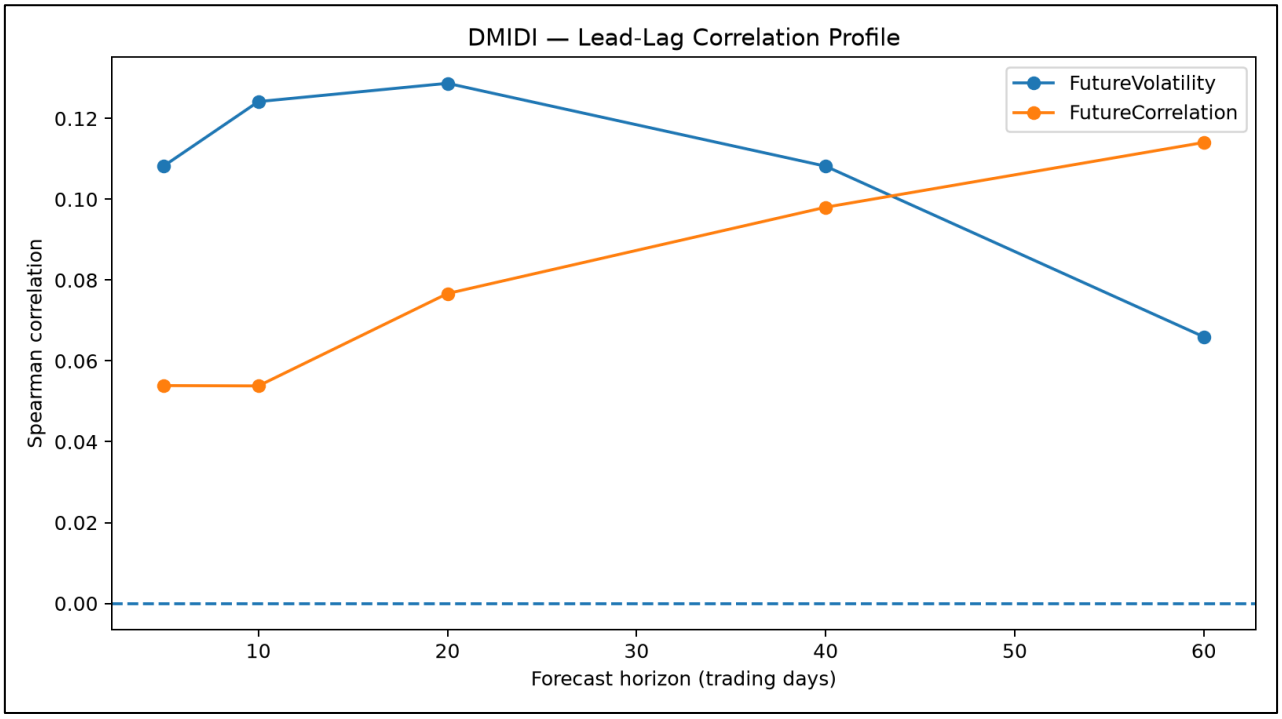
$$CORR_{t+1:t+20} = \alpha + \beta_1 DMIDI_t + \beta_2 PASTCORR_t + \beta_3 VIXLEVEL_t + \beta_4 PASTRV_t + \varepsilon_t$$

where $CORR_{t+1:t+20}$ denotes average cross-asset correlation over the following 20 trading days and $PASTCORR_t$ captures recent observed correlation.

Initial descriptive analysis suggests that higher DMIDI values are associated with somewhat higher future cross-asset correlation. However, the relationship is not strictly linear, motivating the nonlinear analysis developed in the next section.

Across several forward horizons, the rank correlation between DMIDI and future cross-asset correlation is positive and tends to increase with the forecast horizon. This suggests that structural identity drift may contain information about how strongly assets subsequently co-move, even though the relationship is not sufficiently strong to be interpreted as a direct forecasting rule.

Figure 7. Lead-Lag Relationship Between DMIDI and Future Market Conditions



Note: The figure reports rank correlations between current DMIDI and selected future market variables across several forward horizons. The relationships are descriptive and should not be interpreted as causal.

The evidence is more encouraging for future correlation than for future volatility, but the distinction between statistical association and robust predictive performance remains essential. A positive historical relationship does not imply that *DMIDI* systematically improves forecasting accuracy in unseen data.

The shape of the relationship also appears nonlinear. Intermediate and moderately high levels of structural drift are associated with stronger subsequent co-movement, while the relationship does not continue to increase indefinitely. This motivates the quadratic specifications examined next.

5.4 Nonlinear Effects

The previous results suggest that the information contained in structural drift may not be adequately represented by a purely linear specification. In particular, the relationship between DMIDI and subsequent cross-asset correlation appears to change across different levels of structural drift.

To capture this possibility, the analysis introduces nonlinear specifications based on a squared DMIDI term and complementary regime-based comparisons.

5.4.1 Quadratic DMIDI Specification

A quadratic specification allows the marginal effect of DMIDI to vary with the level of structural drift. The forecasting equation for future correlation can be written as:

$$\boldsymbol{CORR_{t+1:t+20} = \alpha + \beta_1 DMIDI_t + \beta_2 DMIDI_t^2 + \beta_3 PASTCORR_t + \beta_4 VIXLEVEL_t + \beta_5 PASTRV_t + \varepsilon_t}$$

The inclusion of $DMIDI_t^2$ allows the relationship to be curved rather than imposing a constant effect across all levels of structural drift.

In the full-sample regression for future cross-asset correlation, both the linear and quadratic DMIDI terms are statistically significant. The linear coefficient is positive, while the squared term is negative, producing an **inverse-U relationship**.

The estimated turning point occurs at a DMIDI level of approximately **8.74**. Below this level, higher structural drift is associated with higher subsequent cross-asset correlation. Beyond the turning point, the estimated relationship begins to flatten and eventually decline.

This turning point should not be interpreted as a structural threshold or trading rule. It is an empirical feature of the fitted historical relationship and must be evaluated out of sample before stronger conclusions can be drawn.

In the quadratic specification, the estimated DMIDI coefficient is approximately **+0.127** and the squared coefficient approximately **−0.0073**, with both terms statistically significant at conventional levels.

The nonlinear result for correlation contrasts with the volatility analysis. Although descriptive volatility plots also appeared nonlinear, the corresponding linear and squared DMIDI coefficients were not statistically significant once conventional volatility controls were included. The evidence for nonlinearity is therefore substantially stronger for future cross-asset correlation than for future realised volatility.

5.4.2 Regime and Decile Analysis

To complement the parametric quadratic regression, the analysis also examines future outcomes across different levels of DMIDI. Observations are grouped into low, middle and high structural-drift regimes, allowing future volatility, correlation and returns to be compared without imposing a specific functional form.

Table 7. Future Market Outcomes Across DMIDI Regimes

DMIDI Regime	Observations	Mean DMIDI	Future Volatility	Future Correlation
Low	472	5.58	0.151	0.141
Middle	2,923	8.68	0.147	0.202
High	751	11.40	0.162	0.175

Note: Regime statistics are descriptive sample averages. They do not control for VIX, recent volatility, recent correlation or other explanatory variables.

The regime averages reinforce the nonlinear pattern observed for future correlation. Average future correlation is lowest in the low-DMIDI regime, increases substantially in the middle regime, and declines again in the high-DMIDI regime.

Future volatility, by contrast, does not display the same clear pattern. Although the high-DMIDI regime has the largest average future volatility, formal regression tests show that these differences do not remain statistically significant once standard risk controls are included.

The in-sample results therefore identify a potentially meaningful nonlinear relationship between structural drift and future cross-asset correlation. However, statistical significance within the estimation sample is not sufficient to establish forecasting value. The next section evaluates whether this relationship survives genuine out-of-sample testing.

5.5 True Out-of-Sample Validation

The in-sample evidence suggests that the relationship between DMIDI and future cross-asset correlation may be nonlinear. However, a relationship identified using the full historical sample may reflect overfitting, data-specific patterns or information that would not have been available in real time.

The analysis therefore evaluates the nonlinear specification under two genuinely out-of-sample designs. The first uses a fixed training period from 2015 to 2021 and a holdout period from 2022 to September 2026. The second uses an expanding-window procedure in which the model is repeatedly re-estimated using only information available prior to each evaluation year.

5.5.1 2015–2021 Training and 2022–2026 Test

The first validation exercise estimates the forecasting models using observations from **2015–2021** and evaluates their performance over the **2022–2026** holdout period.

The benchmark specification uses recent cross-asset correlation, the VIX level and past realised volatility. DMIDI is then added first linearly and subsequently through both linear and quadratic terms.

To reduce numerical dependence between the linear and squared terms, DMIDI is centred around its training-sample mean before constructing the quadratic specification. This transformation does not change the fitted nonlinear relationship but improves the interpretation and numerical stability of the regression coefficients.

Table 8. Holdout Forecast Performance for Future Cross-Asset Correlation

Model	RMSE	MAE	OOS R^2
Past correlation only	0.1585	0.1237	—
Baseline controls	0.1592	0.1248	—
Baseline + DMIDI	0.1595	—	-0.004
Baseline + DMIDI + DMIDI^2	0.1574	0.1234	+0.022

Note: The holdout period runs from 2022 to September 2026. OOS R^2 values compare the Market DNA specification with the baseline control model. Positive values indicate lower forecast error than the benchmark.

The linear DMIDI specification does not improve out-of-sample performance. Its OOS \(R^2\) is slightly negative, indicating forecast accuracy broadly similar to but marginally weaker than the benchmark.

The quadratic DMIDI specification performs somewhat better. Its RMSE declines to approximately **0.1574**, compared with **0.1592** for the baseline model, producing an OOS \(R^2\) of approximately **+2.2%**.

This represents a modest improvement rather than a large forecasting gain. A forecast-error comparison produces a p-value of approximately **0.085**, providing suggestive but not conventionally strong statistical evidence that the nonlinear specification reduces prediction error.

5.5.2 Expanding-Window Validation

A single train-test split may depend excessively on the selected dates. To assess the stability of the result, the analysis therefore performs an expanding-window validation.

For each evaluation year, the model is estimated using all eligible observations available before that year. The estimation sample expands progressively through time, while future observations remain excluded from model fitting. A 20-trading-day purge is applied around the forecasting boundary to reduce potential overlap between the training information and the forward target.

Table 9. Expanding-Window OOS R^2 by Evaluation Year

Year	OOS R^2
2018	-0.100
2019	+0.044
2020	-0.079
2021	+0.046
2022	+0.039
2023	-0.049
2024	+0.040
2025	+0.052
2026	+0.029

Note: Positive OOS R^2 values indicate lower forecast error for the quadratic DMIDI model relative to the corresponding benchmark specification.

The yearly results are mixed but not random in sign. The quadratic DMIDI specification improves forecast accuracy in **six of the nine evaluation years**, with a median yearly OOS R^2 of approximately **+3.9%**.

When all expanding-window forecasts are pooled, the baseline model produces an RMSE of approximately **0.1355**, while the quadratic DMIDI specification achieves an RMSE of approximately **0.1348**. The corresponding pooled OOS R^2 is approximately **+1.1%**.

However, the pooled forecast-error improvement is not statistically significant, with a p-value of approximately **0.48**. The evidence therefore suggests a small and recurrent information contribution, but not a sufficiently robust improvement to support a strong forecasting claim.

5.5.3 Interpretation of Forecasting Results

The forecasting evidence leads to a more nuanced interpretation of Market DNA than initially anticipated.

First, structural identity drift does not improve forecasts of future realised volatility. Both fixed MIDI and dynamic DMIDI underperform simple volatility benchmarks dominated by the VIX.

Second, the evidence for future cross-asset correlation is more encouraging. The nonlinear DMIDI specification produces small out-of-sample improvements in the 2022–2026 holdout sample and positive annual OOS performance in six of nine expanding-window evaluation years.

Third, these gains remain economically modest and statistically uncertain. The pooled expanding-window improvement is small and not statistically significant. Market DNA should therefore not be presented as a superior forecasting model for future correlation.

Instead, the results suggest that structural identity drift may contain information about changes in cross-asset dependence that is not fully captured by conventional market-risk indicators. This information appears nonlinear and episodic rather than stable enough to support a standalone forecasting rule.

The forecasting analysis therefore changes the interpretation of the project. Market DNA is better viewed as a **structural market-state framework** than as a direct prediction engine.

> **Key Forecasting Result**
>
> Market DNA does not improve out-of-sample volatility forecasting. Nonlinear DMIDI contains modest information about future cross-asset correlation, but the improvement is not sufficiently strong or statistically robust to claim forecasting superiority.

Since the strongest contribution of Market DNA lies in describing structural market states rather than forecasting a single risk variable, the next chapter combines drift magnitude and drift breadth into an interpretable market-regime framework.

6. Market DNA Regime Framework

The previous chapters show that structural identity drift is multidimensional and that its strongest contribution lies in describing market structure rather than directly forecasting volatility. The Market DNA framework therefore combines two complementary dimensions: the magnitude of structural drift, measured by DMIDI, and the distribution of that drift across assets, measured by Drift Breadth.

Together, these dimensions provide an intuitive regime framework that distinguishes between stable environments, concentrated asset-specific transformations and broader cross-asset structural shifts.

6.1 Regime Dimensions

The regime framework is built around two variables. The first, **DMIDI**, measures the average multivariate displacement of the six assets from their respective rolling historical Financial DNA structures. The second, **Drift Breadth**, measures how evenly this displacement is distributed across the asset universe.

These dimensions provide different information. A high DMIDI indicates that current factor exposures are unusual relative to recent history, while high Drift Breadth indicates that structural change is broadly distributed across assets. Neither measure alone is sufficient to characterise the nature of the market transition.

Combining the two measures allows episodes with similar aggregate drift magnitude to be distinguished according to their cross-asset structure.

6.2 Percentile-Based Regime Map

Raw DMIDI and Drift Breadth values are transformed into percentiles to make the two dimensions directly comparable. The reference distribution is constructed using the **2015–2021 development period**, which is kept fixed when evaluating later observations.

For each date, the DMIDI percentile indicates the proportion of reference-period observations with a lower DMIDI value. The Drift Breadth percentile is constructed analogously.

This percentile transformation has two advantages. First, it places both indicators on the same 0–100 scale despite their different units. Second, it provides an economically intuitive interpretation of how unusual the current observation is relative to the historical development environment.

Importantly, percentile values describe relative historical position rather than absolute risk. A DMIDI percentile of 90, for example, means that structural drift is higher than approximately 90% of observations in the reference distribution; it does not imply that market risk is 90% or that losses are imminent.

6.3 Regime Classification

The regime map divides the percentile space using the median of each reference distribution. Observations below the 50th percentile are classified as relatively low, while observations at or above the 50th percentile are classified as relatively high.

The resulting four quadrants describe distinct configurations of structural market change.

6.3.1 Stable / Local

Stable / Local corresponds to low DMIDI and low Drift Breadth. Structural displacement is limited and relatively concentrated, indicating that the cross-asset system remains broadly close to its recent historical structure.

6.3.2 Broad Mild Transition

Broad Mild Transition combines low DMIDI with high Drift Breadth. Structural deviations are not individually extreme, but they are distributed relatively broadly across the asset universe. This regime may therefore indicate a gradual or diffuse reconfiguration rather than an acute structural break.

6.3.3 Concentrated Shift

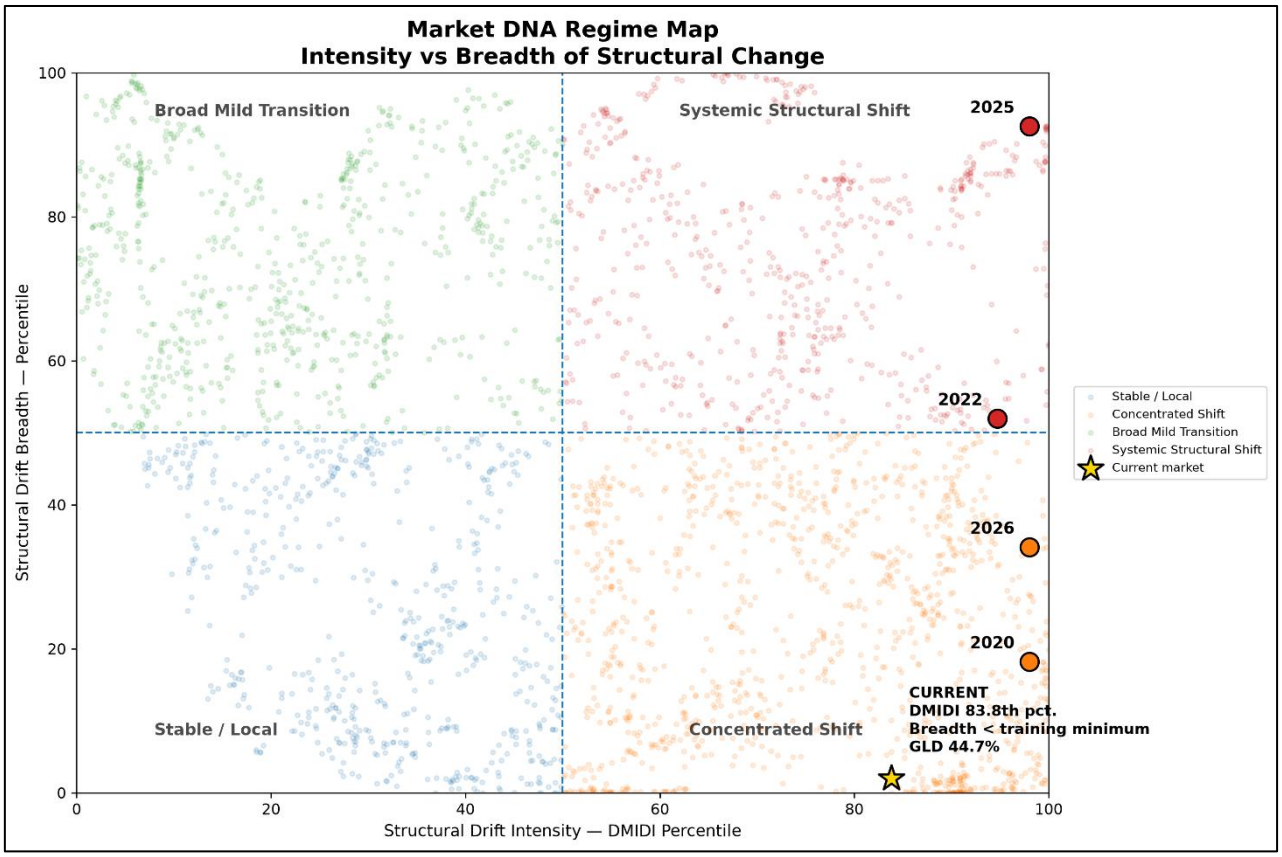
Concentrated Shift combines high DMIDI with low Drift Breadth. Structural drift is elevated, but a disproportionate share of the signal originates from a limited number of assets. The market is therefore experiencing significant structural change without a correspondingly broad cross-asset transformation.

6.3.4 Systemic Structural Shift

Systemic Structural Shift combines high DMIDI with high Drift Breadth. Structural displacement is both elevated and broadly distributed, indicating that several assets are simultaneously experiencing unusual changes in their Financial DNA.

The regime labels are descriptive rather than predictive. In particular, a Systemic Structural Shift should not automatically be interpreted as a crisis, while a Stable / Local regime does not guarantee low subsequent portfolio risk.

Figure 8. Market DNA Regime Map



Note: DMIDI and Drift Breadth are expressed as percentiles relative to the 2015–2021 reference distribution. The horizontal and vertical boundaries correspond to the 50th percentiles of the two reference distributions. Historical episodes are highlighted for interpretation.

Figure 8 shows that structurally important episodes can occur through different combinations of drift magnitude and breadth. Some episodes are characterised by broadly distributed cross-asset changes, while others are driven primarily by a narrower group of assets. The regime framework therefore provides information that would be lost if DMIDI were considered in isolation.

6.4 Current Market DNA State

At the end of the sample, on 30 September 2026, DMIDI stands at approximately 10.71, corresponding to the 83.8th percentile of the 2015–2021 reference distribution. Structural drift is therefore clearly elevated relative to the historical development period.

Drift Breadth is approximately 0.847. Despite appearing relatively high in absolute terms, this value lies below the minimum Breadth observation recorded in the 2015–2021 reference distribution and therefore corresponds to a reference percentile of approximately 0.0.

The combination of elevated DMIDI and exceptionally low relative breadth places the current observation in the **Concentrated Shift** regime.

The asset-level decomposition confirms this interpretation. GLD is the dominant contributor to current structural drift, accounting for approximately **44.7%** of total DIDS. The current signal is therefore not best interpreted as a uniform market-wide transformation, but as an elevated structural shift concentrated disproportionately in a smaller part of the cross-asset universe.

Current Market DNA State — 30 September 2026

DMIDI: 10.71
DMIDI percentile: 83.8
Drift Breadth: 0.847
Breadth percentile: below the 2015–2021 training minimum
Dominant contributor: GLD
Dominant contribution: 44.7%
Regime: Concentrated Shift

The regime map converts the Market DNA indicators into a compact structural interpretation, but aggregate regime labels alone do not explain the economic nature of individual episodes. The next chapter therefore examines selected historical cases in greater detail, combining DMIDI, Drift Breadth, asset contributions and underlying factor-exposure changes.

7. Historical Case Studies

The aggregate Market DNA indicators identify when structural change is unusually large and whether it is concentrated or broadly distributed. However, the same aggregate regime can arise from very different underlying changes in asset-level factor exposures.

This chapter therefore examines four historical episodes in greater detail: 2020, 2022, 2025 and 2026. The objective is not to attribute market events causally to the Market DNA indicators, but to understand how different structural configurations appear within the framework.

For each case, the analysis combines the DMIDI and Drift Breadth percentiles, the distribution of asset-level contributions, and the standardized rolling-beta deviations of the dominant asset. Within each selected year, the episode date is determined mechanically from the structural-drift measures rather than chosen solely from visual inspection.

7.1 2020 — Concentrated Post-Shock Transformation

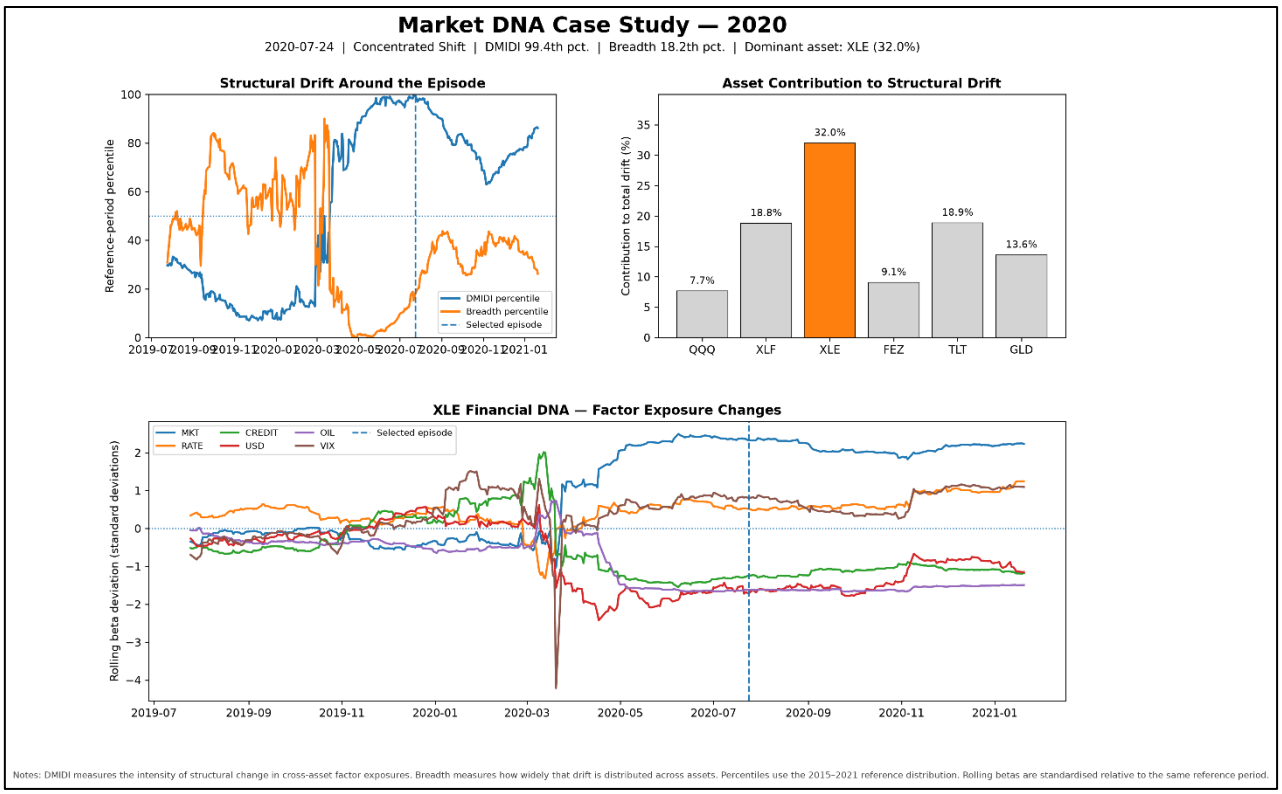
The 2020 episode illustrates a **large but concentrated structural transformation**. On 24 July 2020, DMIDI reaches approximately the 99.4th percentile of the reference distribution, while Drift Breadth remains much lower at approximately the 18.2nd percentile.

The episode is therefore classified as a Concentrated Shift. XLE is the largest contributor, accounting for approximately **32.0%** of total structural drift, while the remaining contribution is distributed across the other assets at substantially lower individual levels.

The underlying XLE factor exposures show why the multivariate drift measure becomes elevated. Relative to the 2015–2021 reference distribution, the market beta moves strongly upward, while credit, US-dollar and oil sensitivities shift materially in the opposite direction. The resulting Financial DNA differs substantially from its earlier configuration across several dimensions rather than through a single isolated coefficient.

Importantly, the selected date should not be interpreted as the moment at which the framework "detected" the COVID-19 shock. Financial DNA is estimated using a 252-trading-day rolling window, meaning that large shocks can continue to reshape estimated factor exposures for months after the initial market event. The July 2020 observation is therefore better interpreted as evidence of a **persistent post-shock structural transformation** in the behaviour of the energy sector.

Figure 9. Market DNA Case Study — 2020 Concentrated Post-Shock Transformation



Note: The upper-left panel shows DMIDI and Drift Breadth percentiles relative to the 2015–2021 reference distribution. The upper-right panel reports asset contributions to total structural drift. The lower panel shows standardized deviations of XLE rolling factor betas from their reference-period distributions.

7.2 2022 — Systemic Structural Transition

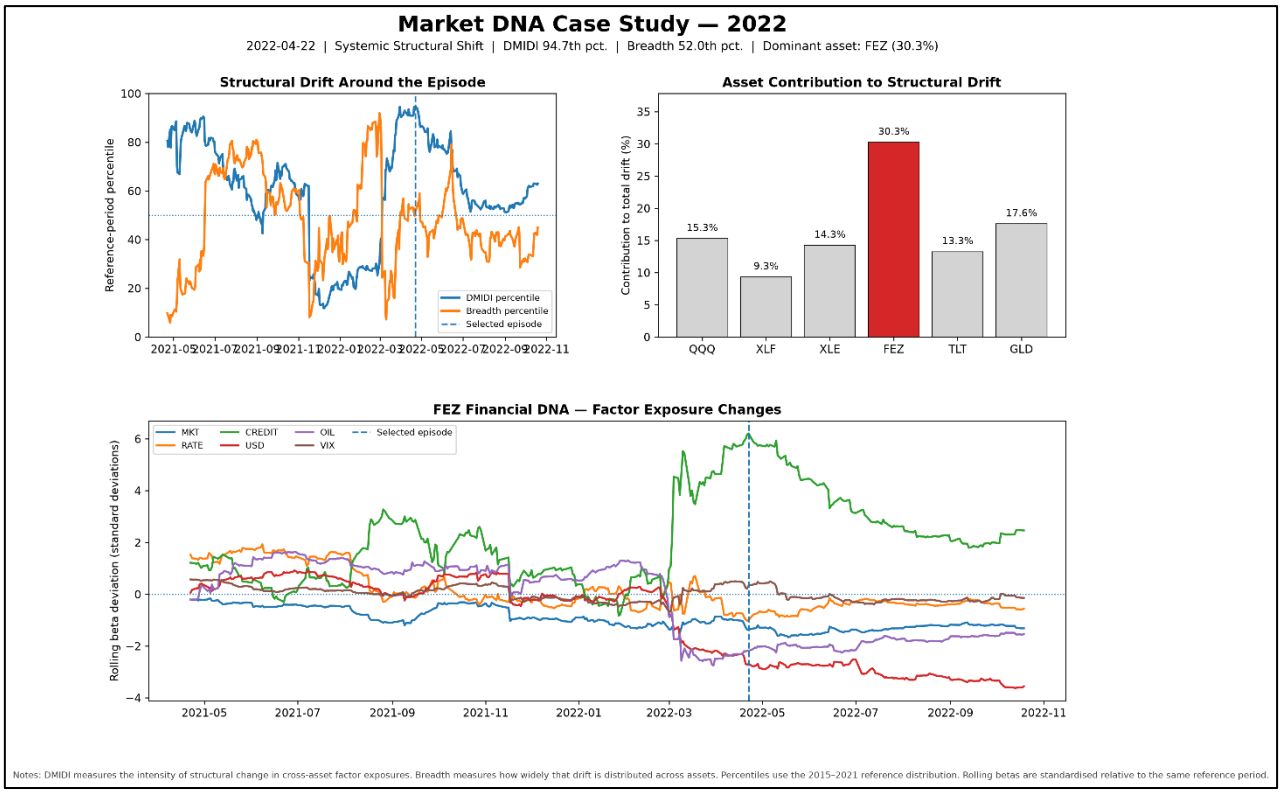
The 2022 episode represents a different configuration. On 22 April 2022, DMIDI is approximately at the **94.7th percentile**, while Drift Breadth reaches approximately the **52.0th percentile**. Both dimensions therefore lie above their respective regime boundaries, placing the observation in the Systemic Structural Shift quadrant.

FEZ is the largest individual contributor, accounting for approximately **30.3%** of total structural drift. However, meaningful contributions are also observed across the other assets, consistent with a broader structural transition than in the 2020 case.

The FEZ factor-exposure decomposition reveals particularly large changes in credit sensitivity. The standardized credit beta rises to roughly six historical standard deviations above its reference-period mean around the selected episode. At the same time, several other exposures, including US-dollar, oil and market sensitivities, move materially away from their historical configurations.

The episode therefore illustrates how a Systemic Structural Shift can combine one particularly important asset-level transformation with simultaneous structural displacement elsewhere in the cross-asset universe. The regime classification does not require identical changes across all assets; rather, it reflects the coexistence of elevated aggregate drift and sufficiently broad participation.

Figure 10. Market DNA Case Study — 2022 Systemic Structural Transition



Note: Factor-exposure deviations are standardized relative to the 2015–2021 reference distribution. They measure how unusual the estimated rolling betas are and should not be interpreted as standardized asset returns.

7.3 2025 — Broad Systemic Structural Shift

The 2025 episode provides the clearest example of a broad systemic transformation within the Market DNA framework. On 13 June 2025, DMIDI reaches approximately the **99.8th percentile**, while Drift Breadth reaches the **92.6th percentile** of the reference distribution.

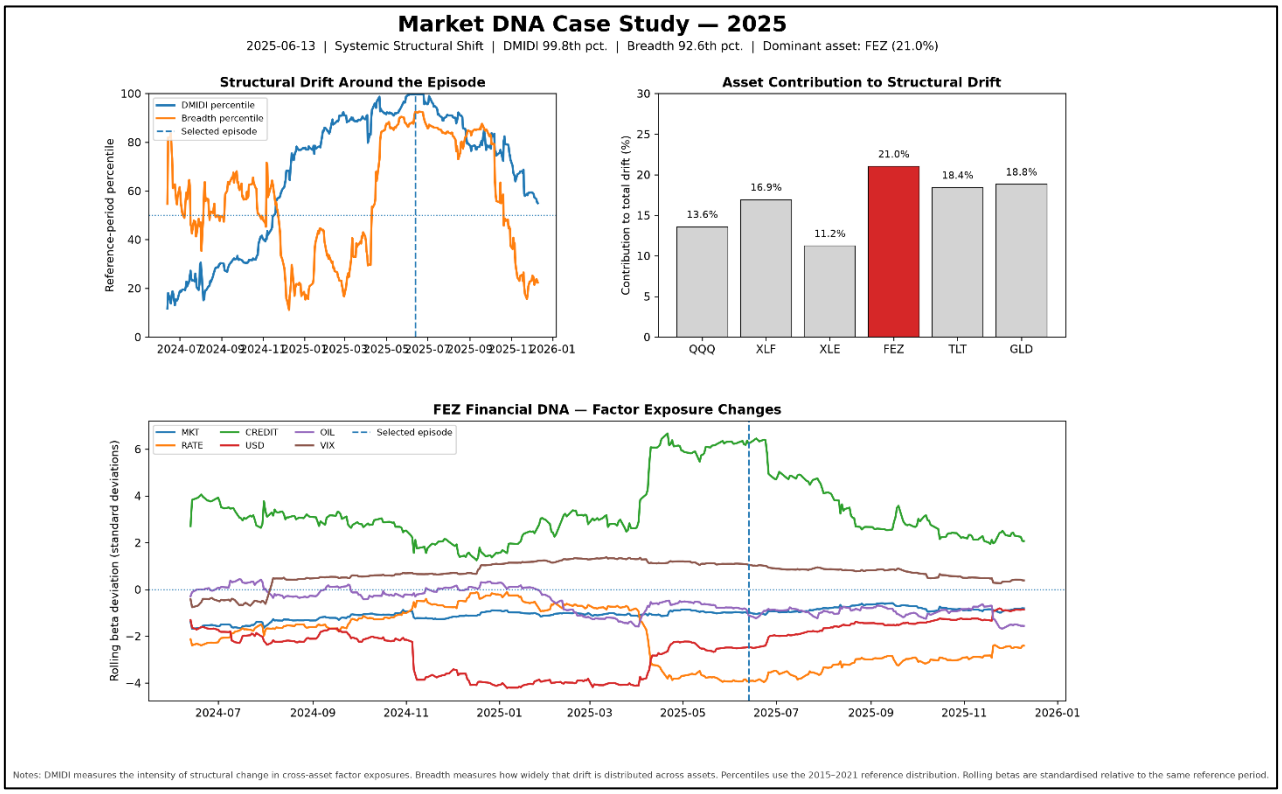
Unlike the concentrated episodes, no single asset dominates the aggregate signal. FEZ is the largest contributor at approximately **21.0%**, followed closely by GLD and TLT, while the remaining assets also contribute materially. The relatively balanced attribution explains the exceptionally high Drift Breadth.

The FEZ Financial DNA decomposition nevertheless shows that important asset-specific transformations remain present within the broader regime. Credit sensitivity is approximately six standard deviations above its reference-period level, while the interest-rate beta is strongly negative relative to its historical distribution. US-dollar exposure is also materially below its reference level.

The significance of the episode therefore lies not simply in the extremity of FEZ's own factor changes, but in the simultaneous participation of the broader asset universe. High DMIDI and high Drift Breadth together identify a market environment in which structural change is both **large and widely distributed**.

Among the historical cases considered here, 2025 provides the most direct illustration of what the framework defines as a Systemic Structural Shift.

Figure 11. Market DNA Case Study — 2025 Broad Systemic Structural Shift



Note: The high Drift Breadth percentile reflects the relatively balanced distribution of structural drift across the six assets. FEZ is the largest contributor but represents only approximately one fifth of total drift.

7.4 2026 — Concentrated Gold-Led Drift

The 2026 case illustrates another form of concentrated structural change. On 30 March 2026, DMIDI reaches approximately the **99.7th percentile**, while Drift Breadth remains at approximately the **34.1st percentile**. The episode is therefore classified as a Concentrated Shift.

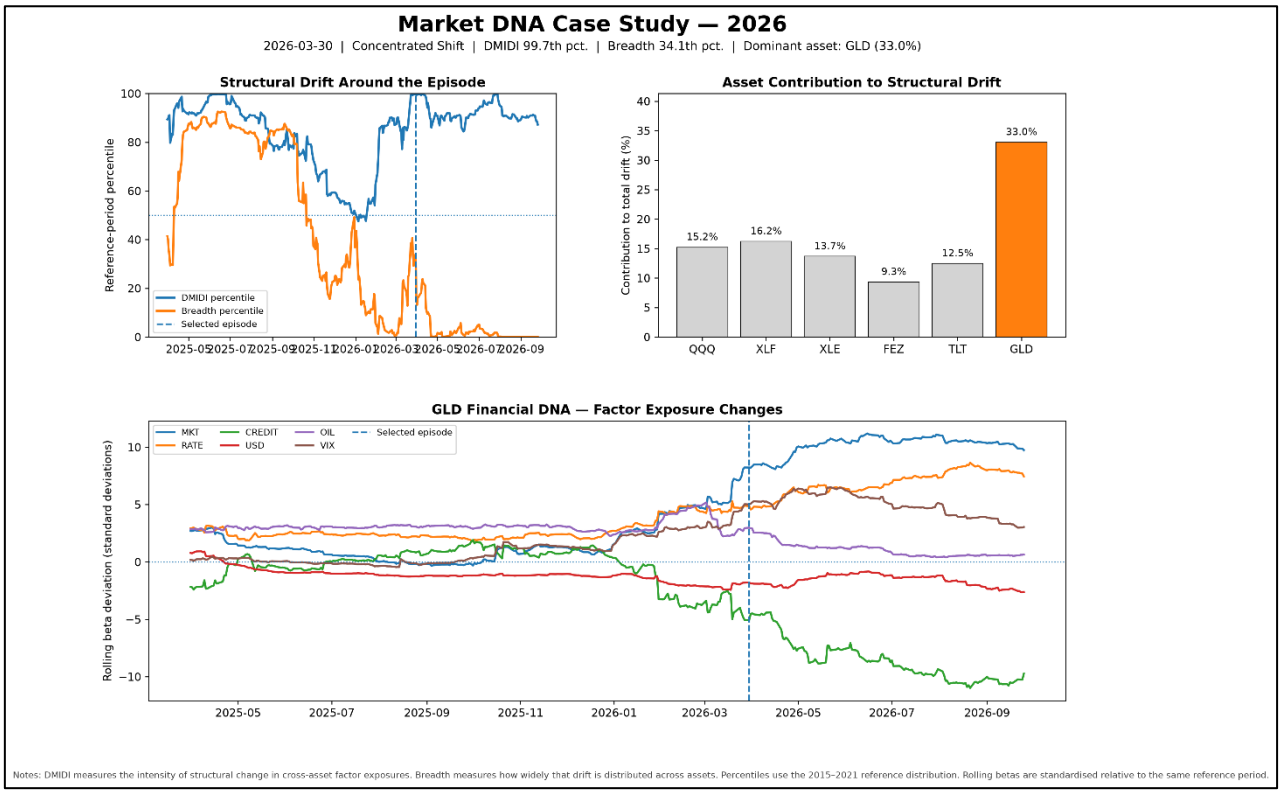
GLD is the dominant contributor, accounting for approximately **33.0%** of total structural drift. The remaining five assets each contribute materially less, producing a relatively concentrated cross-asset structure despite the extreme aggregate DMIDI level.

The GLD Financial DNA decomposition reveals unusually large movements across several factor sensitivities. Market and interest-rate exposures move substantially above their historical reference levels, while credit sensitivity moves strongly in the opposite direction. US-dollar and volatility-related sensitivities also undergo material changes.

These movements illustrate why the Mahalanobis framework is useful. The structural signal does not depend on one beta crossing a predefined threshold; it reflects an unusual multivariate configuration in which several relationships jointly move away from their recent historical covariance structure.

The case also demonstrates why high DMIDI should not automatically be interpreted as a market-wide systemic event. Despite extremely elevated structural displacement, the relatively low Breadth percentile indicates that a disproportionate share of the transformation is concentrated in GLD rather than evenly distributed across the full asset universe.

Figure 12. Market DNA Case Study — 2026 Concentrated Gold-Led Drift



Note: Large standardized beta deviations indicate that GLD's estimated factor sensitivities are far from their historical beta distributions. They do not represent ten-standard-deviation asset returns or direct measures of price risk.

The four case studies demonstrate that similar levels of aggregate structural drift can correspond to very different market configurations. The 2020 and 2026 episodes are characterised by extreme but relatively concentrated transformations, while 2022 crosses into a broader structural regime and 2025 exhibits exceptionally widespread cross-asset participation.

Market DNA therefore adds information not only through the magnitude of DMIDI, but through the joint interpretation of drift magnitude, breadth, asset attribution and the underlying factor-exposure changes. These structural distinctions provide the basis for the portfolio-risk application developed in the next chapter.

8. Portfolio Risk Application: Scenario-Conditioned Monte Carlo

The Market DNA framework is primarily designed to identify and interpret changes in financial-market structure. A natural extension is to examine whether these structural states can also be translated into economically meaningful portfolio-risk scenarios.

This chapter therefore develops a **scenario-conditioned Monte Carlo simulation**. Rather than assuming normally distributed returns or estimating a parametric return model, the simulation resamples historical multi-asset return blocks associated with different Market DNA environments.

The objective is not to forecast the future distribution of returns with certainty. Instead, the exercise illustrates how portfolios might behave under historical return dynamics associated with structurally different market states.

8.1 Motivation

Traditional Monte Carlo models often impose distributional assumptions on asset returns. In contrast, the Market DNA risk application uses an empirical block-bootstrap approach so that simulated paths remain linked to historical cross-asset behaviour.

Conditioning the historical return blocks on structural Market DNA states makes it possible to compare the portfolio distributions associated with three environments: Stable / Local states, states resembling the current Market DNA configuration, and Systemic Structural Shifts.

This approach connects the structural indicators developed in the previous chapters with familiar risk-management measures such as Value at Risk, Expected Shortfall, loss probabilities and maximum drawdown.

8.2 Scenario Construction

Historical observations are divided into three scenario sets according to their Market DNA characteristics. The objective is to preserve economically interpretable structural conditions rather than to rank the scenarios mechanically from low to high risk.

8.2.1 Stable / Local States

Stable / Local observations correspond to periods characterised by relatively low structural drift and low Drift Breadth. These observations represent market environments in which the cross-asset factor structure remains comparatively close to its recent historical configuration.

8.2.2 Current-Like Analogues

Current-like observations are selected by identifying historical Market DNA states closest to the current configuration in the two-dimensional DMIDI-percentile and Drift-Breadth-percentile space. The objective is to construct an empirical scenario set resembling the structural characteristics observed at the end of the sample rather than simply using observations belonging to the same quadrant.

The current-like sample is comparatively small and historically concentrated. After the non-overlapping block filter is applied, only **48 eligible five-day blocks** remain, and a substantial share of the underlying analogues originates from 2026. Results for this scenario should therefore be interpreted with greater caution.

8.2.3 Systemic Shift States

Systemic Shift observations combine elevated DMIDI with broad structural participation. These states correspond to periods in which several assets simultaneously exhibit unusual changes in their Financial DNA.

8.3 Block-Bootstrap Simulation

Portfolio outcomes are simulated over a **20-trading-day horizon** using historical blocks of **five consecutive trading days**. Each Monte Carlo path therefore combines four empirical five-day blocks drawn from the relevant scenario distribution.

A block-bootstrap procedure is preferred to independent daily resampling because it preserves short-run temporal structure within each block as well as the contemporaneous dependence between the six asset returns.

To reduce excessive duplication of nearly identical historical episodes, the final specification uses **non-overlapping historical blocks** as its primary input. This produces 133 eligible blocks for the Stable / Local scenario, 48 for the Current-like scenario and 136 for the Systemic Shift scenario.

8.3.2 Portfolio Construction

The simulated portfolio contains QQQ, XLF, XLE, FEZ, TLT and GLD with equal weights. Equal weighting is deliberately used to keep the risk exercise transparent and to prevent the results from depending on an additional portfolio-optimisation model.

Within each historical block, the simultaneous returns of all six assets are retained. The simulated scenarios therefore preserve the empirical cross-asset relationships observed under the corresponding Market DNA states.

8.4 Portfolio Risk Metrics

The simulated distributions are evaluated using complementary measures of terminal and path-dependent risk. Value at Risk (VaR) reports the loss threshold exceeded with a specified probability, while Expected Shortfall (ES) measures the average loss conditional on exceeding that threshold. The analysis reports both 95% and 99% confidence levels.

Loss probabilities measure the frequency of simulated 20-day returns below selected negative thresholds. Maximum drawdown complements these terminal measures by capturing the largest peak-to-trough decline experienced within each simulated path.

8.4.1 Value at Risk

Value at Risk (VaR) measures the loss threshold that is exceeded with a given probability over the simulation horizon. In this project, VaR is evaluated over the simulated 20-trading-day portfolio return distribution at both the 95% and 99% confidence levels.

For example, a 99% VaR of 8% indicates that only approximately 1% of simulated portfolio outcomes produce a loss larger than 8% over the 20-day horizon.

VaR provides an intuitive measure of downside exposure, but it does not describe the severity of losses once the threshold has been exceeded. For this reason, it is complemented by Expected Shortfall.

8.4.2 Expected Shortfall

Expected Shortfall (ES) measures the average loss conditional on the portfolio return falling beyond the corresponding VaR threshold. It therefore provides information about the severity of extreme downside outcomes rather than only identifying the boundary of the tail.

The analysis reports ES at both the 95% and 99% levels. Because Expected Shortfall incorporates the magnitude of losses beyond VaR, it is particularly useful when comparing scenarios with different tail shapes.

VaR and Expected Shortfall are reported as **positive loss magnitudes** throughout the analysis. Higher values therefore indicate more severe downside risk.

8.4.3 Loss Probabilities

The simulation also reports the probability that the 20-day portfolio return falls below predefined loss thresholds. Two thresholds are considered: a loss greater than **5%** and a loss greater than **10%**.

These probabilities complement VaR and Expected Shortfall by providing a direct frequency-based interpretation of adverse outcomes. They are particularly useful when two scenarios have similar tail thresholds but different probabilities of experiencing moderate or severe losses.

8.4.4 Maximum Drawdown

Terminal portfolio returns do not capture the full path experienced by an investor. A portfolio may finish a 20-day period with a relatively moderate gain or loss while having experienced a substantial temporary decline during the simulation.

Maximum drawdown therefore measures the largest peak-to-trough decline within each simulated 20-day path. The analysis reports the average maximum drawdown across the simulated paths for each Market DNA scenario.

This path-dependent measure complements VaR, Expected Shortfall and terminal loss probabilities by capturing temporary portfolio stress that may not be visible from the final return alone.

Together, these metrics provide complementary views of portfolio risk: VaR identifies a loss threshold, Expected Shortfall measures the severity of extreme losses, loss probabilities quantify the frequency of adverse outcomes, and maximum drawdown captures path-dependent stress.

8.5 Monte Carlo Results

The scenario-conditioned simulations produce materially different portfolio-return distributions, but the results do not follow a simple ranking from structurally stable to structurally extreme market states.

Table 10 summarises the main results obtained from 10,000 simulated 20-day paths for each scenario using non-overlapping five-day historical return blocks.

Table 10. Scenario-Conditioned Monte Carlo Results

Metric	Stable / Local	Current-like	Systemic Shift
Eligible historical blocks	133	48	136
Mean 20-day return	0.49%	1.72%	0.71%
Median 20-day return	0.84%	1.49%	1.08%
Return volatility	3.51%	3.83%	3.26%
VaR 95%	5.01%	4.20%	5.53%
VaR 99%	12.81%	6.63%	8.61%
ES 95%	9.80%	5.67%	7.49%
ES 99%	14.24%	7.74%	10.35%
P(Return < −5%)	5.02%	3.06%	6.09%
P(Return < −10%)	2.64%	0.03%	0.43%
Mean maximum drawdown	−2.78%	−3.28%	−2.61%

Note: Results are based on 10,000 simulated 20-day portfolio paths using five-day non-overlapping historical return blocks. VaR and Expected Shortfall are expressed as positive loss magnitudes.

Stable / Local scenario

The Stable / Local scenario produces the most severe extreme left-tail outcomes. Its 99% VaR is approximately **12.8%**, while its 99% Expected Shortfall reaches approximately **14.2%**. The probability of a loss greater than 10% is also the highest of the three scenarios at approximately **2.6%**.

This result should not be interpreted as evidence that structurally stable markets are inherently more dangerous. Historical inspection shows that some Stable / Local observations immediately precede abrupt market shocks, including the March 2020 collapse. A structural indicator can therefore describe the market state immediately before a discontinuous event without anticipating that event.

Current-like scenario

The Current-like scenario exhibits the highest mean and median terminal returns and the most favourable VaR and Expected Shortfall estimates. Its 99% VaR is approximately **6.6%**, substantially below the corresponding Stable / Local estimate.

However, the scenario also produces the largest average maximum drawdown, at approximately **−3.3%**. This illustrates the distinction between terminal and path-dependent risk: favourable final outcomes do not necessarily imply smoother portfolio trajectories.

The Current-like results must also be interpreted cautiously because the historical analogue set is relatively small and concentrated in recent observations.

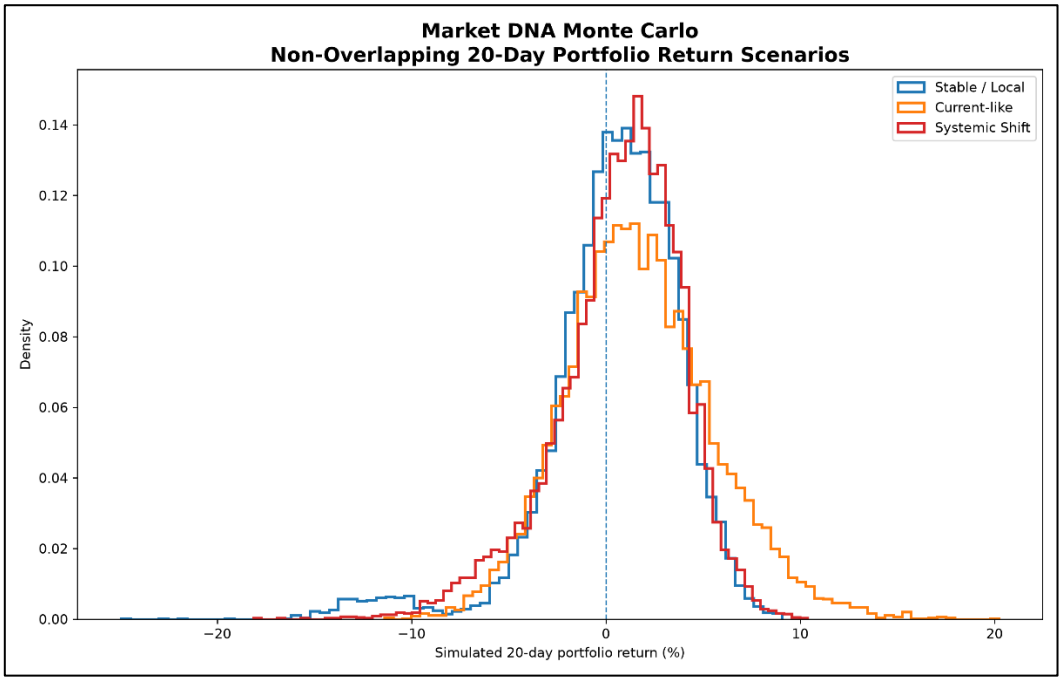
Systemic Shift scenario

The Systemic Shift scenario occupies an intermediate position in terms of extreme tail losses but produces the highest probability of a loss greater than 5%, at approximately **6.1%**.

This suggests that broad structural transformations may be associated with a greater frequency of moderate adverse outcomes without necessarily generating the most severe extreme-tail losses observed in the historical sample.

The Monte Carlo results therefore reject a simple interpretation in which higher structural drift automatically implies higher portfolio risk. Market DNA regimes describe the configuration of factor relationships, while subsequent portfolio outcomes remain influenced by shocks and return dynamics that are not fully captured by the structural indicators.

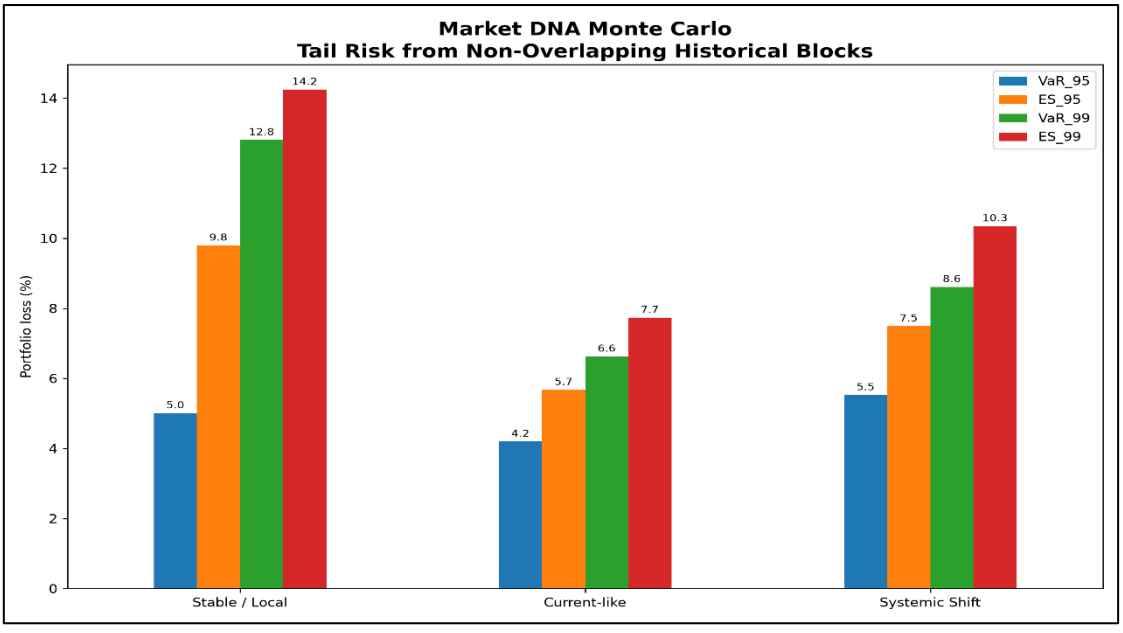
Figure 13. Simulated 20-Day Portfolio Return Distributions by Market DNA Scenario



Note: Each distribution is generated from 10,000 scenario-conditioned block-bootstrap simulations using non-overlapping five-day historical return blocks.

The simulated distributions confirm that the three scenarios differ not only in average outcomes but also in the shape and thickness of their downside tails. In particular, the Stable / Local distribution contains relatively rare but severe negative outcomes, while the Current-like scenario displays a narrower extreme left tail.

Figure 14. Simulated Portfolio Tail Risk by Market DNA Scenario



Note: VaR and Expected Shortfall are expressed as positive portfolio-loss magnitudes. Higher values indicate more severe simulated downside risk.

Figure 14 makes the non-monotonic nature of the results particularly clear. The Systemic Shift scenario does not produce the largest extreme-loss estimates, while the structurally stable scenario exhibits the largest 99% tail-risk measures.

8.6 Robustness Analysis

Because historical block-bootstrap results can be sensitive to the construction of the sampling set, additional robustness checks are performed before interpreting the scenario differences. Particular attention is given to overlapping historical blocks and to the limited historical diversity of the Current-like analogue sample.

8.6.1 Non-Overlapping Historical Blocks

The initial Monte Carlo implementation allowed overlapping five-day historical blocks. This approach increases the number of available observations but can give excessive weight to closely related return sequences from the same market episode.

For example, consecutive signal dates can generate forward blocks containing many of the same trading days. Treating these observations as independent bootstrap building blocks would therefore overstate the effective amount of historical information available.

The robustness analysis consequently reconstructs the scenario samples using **non-overlapping five-day return blocks**. The number of eligible blocks declines from 594 to 133 for Stable / Local observations, from 250 to 48 for Current-like observations, and from 620 to 136 for Systemic Shift observations.

Despite the substantial reduction in sample size, the qualitative conclusions remain broadly unchanged. In particular, the Stable / Local scenario continues to exhibit the heaviest extreme left tail, while Systemic Shift results remain comparatively stable across the alternative sampling procedures.

Non-overlapping blocks are therefore retained as the **primary Monte Carlo specification** reported in the main results.

8.6.2 Concentration of Current-Like Analogues

The Current-like scenario presents a separate robustness concern because the nearest historical analogues to the current Market DNA state are not evenly distributed through time.

Of the original 250 nearest structural analogues, approximately **54.8% originate from 2026**, while 2020 accounts for approximately 23.6%. Only a small number of observations come from several other years. The effective historical diversity of the analogue set is therefore substantially lower than the raw number of observations suggests.

Expressed in terms of concentration, the analogue distribution corresponds to an effective historical coverage of only around **2.7 equally represented years**.

This limitation matters because the simulated Current-like distribution may reflect specific features of recent market conditions rather than a broad collection of independent historical precedents. Its apparently favourable terminal-return and tail-risk profile should therefore not be interpreted as a reliable forecast of the current regime.

8.6.3 Interpretation of Tail Risk

The robustness analysis highlights an important distinction between structural market state and subsequent portfolio risk.

Market DNA is designed to measure changes in factor relationships. It does not directly model the arrival of new information, geopolitical shocks, policy surprises, liquidity events or other discontinuities capable of generating abrupt price movements.

Consequently, a Stable / Local structural state can immediately precede a severe market loss, while an extreme Systemic Structural Shift does not necessarily produce the largest subsequent tail event. The March 2020 observations provide a particularly clear example: several structurally stable signal dates are followed almost immediately by exceptionally negative five-day portfolio returns.

The Monte Carlo exercise should therefore be interpreted as a scenario-conditioned risk application rather than a forecasting model.

Its purpose is to translate different Market DNA configurations into empirical portfolio distributions and to illustrate how terminal returns, tail losses and drawdowns differ across structural environments. It does not establish a deterministic mapping from Market DNA regime to future portfolio performance.

Overall, the portfolio application reinforces the broader conclusion of the project. Market DNA provides information about the **structure of financial relationships**, while structural state and subsequent financial risk remain distinct concepts.

The absence of a monotonic risk ranking is itself informative. It suggests that structural regime indicators are best used as contextual tools for understanding the market environment and conditioning risk scenarios, rather than as standalone predictors of future losses.

9. Conclusion

9.1 Main Findings

This project develops Market DNA as a framework for measuring changes in the economic relationships that define financial assets through time. Rather than characterising an asset only by its return, volatility or conventional asset-class label, the approach represents financial identity through a vector of rolling sensitivities to six common market factors.

The empirical analysis shows that these factor exposures vary materially through time. Mahalanobis distance provides a natural way to measure the joint displacement of these sensitivities while accounting for their historical variability and covariance structure. This produces asset-level Identity Drift Scores and, ultimately, the Dynamic Market Identity Drift Index (DMIDI).

A key result of the project is that aggregate structural drift cannot be interpreted without considering its distribution across assets. Drift Breadth therefore complements DMIDI by distinguishing concentrated structural transformations from broad cross-asset shifts. Historical episodes demonstrate that similarly elevated DMIDI levels can correspond to very different market structures.

The forecasting results also refine the original hypothesis of the project. Market DNA does **not** improve out-of-sample forecasts of future realised volatility relative to conventional benchmarks dominated by the VIX. The evidence for future cross-asset correlation is more encouraging: nonlinear DMIDI specifications produce modest out-of-sample improvements, but the gains are too small and statistically uncertain to support a claim of forecasting superiority.

The regime framework and Monte Carlo application therefore support a broader interpretation of Market DNA. Its primary value lies in identifying and decomposing **structural changes in financial-market relationships**, rather than in predicting the magnitude of future price movements.

9.2 Limitations and Interpretation

Several limitations should be considered when interpreting the results. First, Financial DNA depends on the selected factor model. The six factors used in this project capture major macro-financial transmission channels, but they cannot represent every economic force affecting the assets. Alternative factors or asset-specific models could therefore produce different structural-drift measures.

Second, rolling-window estimates introduce persistence into the factor exposures. A large market shock can influence rolling betas for many months after the original event, meaning that Market DNA should be interpreted as a measure of evolving structural relationships rather than as a real-time shock detector.

Third, the dynamic benchmark depends on the chosen 504-observation reference window and 252-trading-day separation gap. These choices provide a reasonable balance between historical stability and adaptability, but alternative window lengths could affect the timing and magnitude of the resulting drift scores.

Fourth, the predictive evidence remains deliberately modest. The nonlinear relationship with future cross-asset correlation is recurrent across several out-of-sample periods, but the pooled improvement is small and not statistically significant. The framework should therefore not be used as a standalone forecasting or trading signal.

Finally, the Monte Carlo application is dependent on the historical scenarios available in the sample. In particular, Current-like analogues are relatively concentrated in recent observations, while structurally stable states can immediately precede discontinuous shocks. The simulated distributions should therefore be interpreted as conditional historical scenarios rather than forecasts of future portfolio returns.

9.3 What Market DNA Adds

The main contribution of Market DNA is the integration of several quantitative tools into a coherent structural framework.

Rolling factor models provide a dynamic representation of asset behaviour. Mahalanobis distance converts changes in multiple factor sensitivities into interpretable asset-level drift scores. Drift Breadth separates concentrated from systemic transformations, while PCA and cross-asset correlations reveal whether structural changes share a common component. The regime map then combines these elements into an intuitive representation of market state.

Importantly, the framework preserves the distinction between **description and prediction**. A market may undergo substantial structural change without subsequently experiencing higher volatility, and a structurally stable environment may still be exposed to an abrupt external shock.

Market DNA therefore adds a layer of information that conventional volatility measures do not provide: it identifies **how the economic relationships underlying asset behaviour are changing**, where those changes originate, and whether they are concentrated or broadly distributed across markets.

In this sense, Market DNA is best understood as a **structural monitoring and risk-context framework**, rather than as a replacement for established forecasting models.

9.4 Possible Extensions

Several extensions could strengthen the framework in future work. The asset universe could be expanded to include additional equity regions, currencies, commodities or credit instruments, while the factor architecture could incorporate inflation expectations, term-premium measures or liquidity indicators.

Alternative estimation methods could also be explored. State-space models, Kalman filtering or time-varying parameter regressions could provide smoother estimates of Financial DNA than fixed rolling windows. Robust covariance estimators could further improve the stability of Mahalanobis distances during periods of limited historical information.

The regime framework could also be tested across alternative reference periods and percentile thresholds, while the predictive analysis could be extended to portfolio correlation, diversification breakdown, drawdown risk or asset-allocation outcomes.

Finally, the scenario-conditioned Monte Carlo framework could be developed into a broader risk-management tool in which portfolio exposures, hedge effectiveness and stress scenarios are dynamically evaluated conditional on the prevailing Market DNA state.

Financial markets rarely change in only one dimension. Prices, volatility and correlations evolve together with the economic relationships that connect assets to common market forces. Market DNA provides a quantitative way to observe this less visible form of change. ***The empirical evidence suggests that structural drift is not a simple predictor of future risk, but it can provide a useful and interpretable perspective on how the architecture of financial markets evolves through time.***

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Technical Appendix

A.1 Data Definitions

The empirical analysis uses daily financial-market data covering the period from 4 January 2010 to 30 September 2026. All series are aligned on common trading dates before model estimation.

The asset universe consists of six liquid exchange-traded instruments representing distinct segments of the financial system: QQQ, XLF, XLE, FEZ, TLT and GLD. The factor architecture contains six common market drivers: a broad equity-market factor, long-term interest-rate changes, credit-market conditions, the US dollar, oil prices and changes in implied market volatility.

Table A1. Data Definitions

Variable	Source / Proxy	Transformation	Role
QQQ	Nasdaq-100 ETF	Daily log return	Asset
XLF	Financial Select Sector ETF	Daily log return	Asset
XLE	Energy Select Sector ETF	Daily log return	Asset
FEZ	Euro Stoxx 50 ETF	Daily log return	Asset
TLT	20+ Year US Treasury ETF	Daily log return	Asset
GLD	Gold ETF	Daily log return	Asset
MKT	VT	Daily log return	Market factor
RATE	DGS10	Daily yield change × 100	Rate factor
CREDIT	HYG − IEF	Difference in daily log returns	Credit factor
USD	UUP	Daily log return	Dollar factor
OIL	WTI crude oil futures	Daily log return	Oil factor
VIX	VIXCLS	Daily change in VIX level	Volatility factor
VIX_LEVEL	VIXCLS	VIX level	Forecast control

Note: RATE is expressed in basis points. VIX_LEVEL is retained as a forecasting control but is not included in the six-dimensional Financial DNA vector.

The main data transformations are defined as follows.

$$r_t = \ln\left(\frac{P_t}{P_{t-1}}\right)$$

where P_t denotes the adjusted closing price at date t. The interest-rate factor is expressed as the daily change in the US 10-year Treasury yield, in basis points:

$$RATE_t = 100(DGS10_t - DGS10_{t-1})$$

Credit conditions are represented by the return differential between high-yield corporate bonds and intermediate US Treasuries:

$$CREDIT_t = r_{HYG,t} - r_{IEF,t}$$

The volatility factor is defined as the daily change in the VIX level:

$$VIX_t = VIXLEVEL_t - VIXLEVEL_{t-1}$$

After transformation, all asset and factor series are merged on common trading dates. Missing observations created by differencing or non-overlapping market calendars are removed before the rolling regressions are estimated.

A.2 Factor Model Specification

For each asset i, daily returns are modelled as a linear function of the six common market factors defined in Appendix A.1. The model is estimated separately for each asset using rolling ordinary least squares (OLS).

The rolling factor model is specified as:

$$\boldsymbol{r_{i,t}} = \boldsymbol{\alpha_{i,t}} + \boldsymbol{\beta_{(MKT,i,t)MKT_t}} + \boldsymbol{\beta_{(RATE,i,t)RATE_t}} + \boldsymbol{\beta_{(CREDIT,i,t)CREDIT_t}} + \boldsymbol{\beta_{(USD,i,t)USD_t}} + \boldsymbol{\beta_{(OIL,i,t)OIL_t}} + \boldsymbol{\beta_{(VIX,i,t)VIX_t}} + \boldsymbol{\varepsilon_{i,t}}$$

where $r_{i,t}$ is the daily return of asset i, $\alpha_{i,t}$ is the rolling intercept, $\beta_{k,i,t}$ is the estimated sensitivity of asset i to factor k at date t, and $\varepsilon_{i,t}$ is the regression residual.

Rolling estimation window

The model is estimated using a **252-trading-day rolling window**, approximately corresponding to one year of daily observations. For each date t, the regression is estimated using the most recent observations contained in the rolling window.

$$\boldsymbol{W_t} = \{\boldsymbol{t - 251}, \dots, \boldsymbol{t}\}$$

A minimum of **200 valid observations** is required for a rolling estimate to be retained. This condition allows the model to tolerate a limited number of missing observations while avoiding estimates based on excessively small samples.

Rolling OLS estimator

Within each rolling window, the coefficient vector is estimated by ordinary least squares:

$$\boldsymbol{\widehat{\beta}_t} = (\boldsymbol{X_t^T X_t})_{\boldsymbol{t}}^{\boldsymbol{(-1)} X_t^T y}$$

where X_t contains the intercept and the six factor observations within the rolling window, while y_t contains the corresponding asset returns.

$$\boldsymbol{\widehat{\theta}_{i,t}} = (\boldsymbol{X_t^T X_t})_{\boldsymbol{i,t}}^{\boldsymbol{(-1)} X_t^T y}$$

where $\hat{\theta}_{i,t}$ contains the estimated intercept and six factor coefficients for asset i

Financial DNA vector

The intercept is not included in the Financial DNA representation. At each date, the Financial DNA of asset i is defined exclusively by the six estimated factor sensitivities:

$$\boldsymbol{\beta_{i,t}} = \left(\boldsymbol{\beta_{MKT,i,t}}, \boldsymbol{\beta_{RATE,i,t}}, \boldsymbol{\beta_{CREDIT,i,t}}, \boldsymbol{\beta_{USD,i,t}}, \boldsymbol{\beta_{OIL,i,t}}, \boldsymbol{\beta_{VIX,i,t}}\right)^{\boldsymbol{T}}$$

The resulting six-dimensional vector is the primary input to the identity-drift calculations developed in Appendix A.3.

The rolling coefficients are interpreted as conditional statistical exposures rather than causal parameters. Changes in a beta indicate that the historical relationship between an asset and a factor has changed within the estimation window; they do not by themselves establish a causal effect of the factor on the asset return.

No forecasting interpretation is attached directly to the rolling betas. Their purpose is to provide a time-varying representation of asset structure that can subsequently be compared with historical reference distributions.

A.3 Mahalanobis Distance Construction

Structural identity drift is measured by comparing the current Financial DNA vector with a historical reference distribution. Because the six factor sensitivities have different variances and may be correlated, the comparison is based on Mahalanobis distance rather than on a simple Euclidean distance.

For asset i, let μ_i denote the historical mean Financial DNA vector and $\sum_i$ the corresponding covariance matrix over the selected reference period.

$$\boldsymbol{\mu_i} = \frac{\mathbf{1}}{\boldsymbol{T}} \sum_{\boldsymbol{t=1}}^{\boldsymbol{T}} \boldsymbol{\beta_{i,t}}$$

where T is the number of Financial DNA observations contained in the reference sample.

$$\boldsymbol{\Sigma_i} = \frac{\mathbf{1}}{\boldsymbol{T-1}} \sum_{\boldsymbol{t=1}}^{\boldsymbol{T}} (\boldsymbol{\beta_{i,t}} - \boldsymbol{\mu_i})(\boldsymbol{\beta_{i,t}} - \boldsymbol{\mu_i})^{\boldsymbol{T}}$$

Fixed Identity Drift Score

In the fixed-reference specification, the current Financial DNA vector is compared with the mean and covariance matrix estimated from the **2011–2014 reference period**.

$$\boldsymbol{IDS_{i,t}} = \sqrt{\left((\boldsymbol{\beta_{i,t}} - \boldsymbol{\mu_i})^{\boldsymbol{T}} \boldsymbol{\Sigma_i^{-1}} (\boldsymbol{\beta_{i,t}} - \boldsymbol{\mu_i})\right)}$$

A larger $IDS_{i,t}$ indicates that the current combination of factor sensitivities is further from the historical reference structure after adjusting for both the variability and covariance of the betas.

$$\boldsymbol{MIDI_t} = \frac{\mathbf{1}}{\boldsymbol{N}} \sum_{\boldsymbol{i=1}}^{\boldsymbol{N}} \boldsymbol{IDS_{i,t}} \quad \boldsymbol{with\ N = 6}$$

The fixed MIDI therefore represents the average multivariate structural distance across the six assets.

Dynamic Identity Drift Score

The dynamic specification replaces the fixed historical benchmark with a rolling reference distribution. For each date t, the reference sample contains the **504 historical Financial DNA observations** ending **252 trading days before** the current observation.

$$\boldsymbol{DIDS_{i,t}} = \sqrt{\left((\boldsymbol{\beta_{i,t}} - \boldsymbol{\mu_{i,t}})^{\boldsymbol{T}} \boldsymbol{\Sigma_{i,t}^{-1}} (\boldsymbol{\beta_{i,t}} - \boldsymbol{\mu_{i,t}})\right)}$$

where $\mu_{i,t}$ and $\sum_{i,t}$ are estimated from the rolling reference sample available for asset i at date t.

$$\boldsymbol{DMIDI_t} = \frac{\mathbf{1}}{\boldsymbol{N}} \sum_{\boldsymbol{i=1}}^{\boldsymbol{N}} \boldsymbol{DIDS_{i,t}}$$

The 252-trading-day separation gap is introduced because each Financial DNA vector is itself estimated from a 252-day rolling regression. Without this gap, the current beta vector and the historical reference observations would share a substantial amount of underlying return data, creating mechanical dependence between the observation being evaluated and its benchmark.

Mahalanobis distance assigns greater importance to deviations that are historically unusual and less importance to movements that frequently occur within the reference distribution. It also recognises the covariance structure of the factor sensitivities: a combination of beta movements that is unusual jointly may receive a high drift score even when no individual coefficient appears extreme in isolation.

The fixed and dynamic specifications therefore answer different questions. Fixed MIDI measures distance from the 2011–2014 structural regime, while DMIDI measures how unusual the current Financial DNA is relative to a more recent but deliberately separated historical benchmark.

A.4 Drift Breadth Formula

Drift Breadth measures how evenly structural identity drift is distributed across the six assets. It is derived from the asset-level Dynamic Identity Drift Scores and is based on normalized Shannon entropy.

For each asset i, the contribution share at date t is defined as:

$$\boldsymbol{w_{i,t}} = \frac{\boldsymbol{DIDS_{i,t}}}{\sum_{j=1}^{N} \boldsymbol{DIDS_{j,t}}}$$

where $N = 6$. By construction, the contribution shares are non-negative and sum to one.

$$\sum_{i=1}^{N} \boldsymbol{w_{i,t}} = \mathbf{1}$$

Drift Breadth is calculated as the normalized entropy of the contribution distribution:

$$\boldsymbol{DRIFT_{BREADTH_t}} = -\frac{\sum_{i=1}^{N} \boldsymbol{w_{i,t}} \ln(\boldsymbol{w_{i,t}})}{\ln(\boldsymbol{N})}$$

Dividing by $\ln(N)$ normalizes the entropy measure so that:

$$\mathbf{0} \leq \boldsymbol{DRIFT_{BREADTH_t}} \leq \mathbf{1}$$

A value close to zero indicates that structural drift is concentrated in a small number of assets, while a value close to one indicates that drift is distributed relatively evenly across the asset universe.

The measure reflects the **distribution** of drift rather than its magnitude. Consequently, a high Drift Breadth value does not imply that structural change is large; it only indicates that the observed drift is broadly distributed. For this reason, Drift Breadth is interpreted jointly with DMIDI.

The combination of DMIDI and Drift Breadth forms the basis of the Market DNA regime framework. High DMIDI with low breadth corresponds to a concentrated structural shift, while high DMIDI with high breadth indicates a broad systemic transformation.

A.5 PCA Methodology

Principal Component Analysis (PCA) is applied to the six asset-level Dynamic Identity Drift Score series in order to assess whether structural drift is dominated by a common cross-asset component or remains largely multidimensional.

Because the DIDS series may differ in scale and dispersion, each series is standardized before the PCA is performed.

$$\boldsymbol{z_{i,t}} = \frac{\boldsymbol{DIDS_{i,t}} - \boldsymbol{\mu_i}}{\boldsymbol{\sigma_i}}$$

where μ_i and $\sum_i$ denote the sample mean and standard deviation of the DIDS series for asset i.

The standardized variables are assembled into a data matrix Z. PCA is then performed on the covariance matrix of the standardized series, which is equivalent to analysing their correlation structure.

$$C = \frac{1}{T-1} Z^T Z$$

The principal components are obtained from the eigenvectors of C, while the associated eigenvalues determine the proportion of total variance explained by each component.

$$EVR_k = \frac{\lambda_k}{\sum_{j=1}^{N} \lambda_j}$$

where λ_k is the eigenvalue associated with principal component k.

In the empirical application, the first principal component explains approximately **25.38%** of the total standardized DIDS variation.

The corresponding first-component loadings are approximately 0.729 for QQQ, 0.659 for XLF, 0.378 for XLE, 0.589 for FEZ, −0.186 for TLT and 0.181 for GLD.

The sign of a principal component is arbitrary. Multiplying an eigenvector and its associated scores by −1 produces an equivalent PCA solution. Interpretation should therefore focus on the relative signs and magnitudes of the loadings rather than on their absolute orientation.

The relatively limited variance explained by the first component indicates that structural identity drift cannot be reduced to a single common market factor. A shared cross-equity component is present, but most of the variation remains distributed across additional dimensions. This supports the interpretation of Market DNA as a **multidimensional structural framework**.

A.6 Regression Specifications

The predictive analysis evaluates whether structural identity drift contains incremental information about future market conditions beyond conventional controls. Two primary dependent variables are considered: future realised volatility and future cross-asset correlation, generally measured over a 20-trading-day horizon.

Unless otherwise stated, the explanatory variables are observed at date \(t\), while the dependent variable is constructed from subsequent market observations. This temporal ordering prevents future information from entering the forecasting regressors.

Future realised volatility

The baseline volatility specification controls for the current VIX level and recent realised volatility. The linear DMIDI specification is:

$$RV_{t+1:t+20} = \alpha + \beta_1 DMIDI_t + \beta_2 VIXLEVEL_t + \beta_3 PASTRV_t + \varepsilon_t$$

where $RV_{t+1:t+20}$ denotes realised volatility over the following 20 trading days, $VIXLEVEL_t$ is the current VIX level and $PASTRV_t$ is recent realised volatility.

A nonlinear specification is estimated by adding a squared DMIDI term:

$$RV_{t+1:t+20} = \alpha + \beta_1 DMIDI_t + \beta_2 DMIDI_t^2 + \beta_3 VIXLEVEL_t + \beta_4 PASTRV_t + \varepsilon_t$$

The quadratic term allows the relationship between structural drift and future volatility to vary across different DMIDI levels rather than imposing a constant marginal effect.

Future cross-asset correlation

Future cross-asset correlation is modelled using recent correlation, the VIX level and past realised volatility as controls. The linear specification is:

$$\boldsymbol{CORR_{t+1:t+20} = \alpha + \beta_1 DMIDI_t + \beta_2 PASTCORR_t + \beta_3 VIXLEVEL_t + \beta_4 PASTRV_t + \varepsilon_t}$$

where $CORR_{t+1:t+20}$ denotes average cross-asset correlation over the subsequent 20 trading days and $PASTCORR_t$ represents recent cross-asset correlation.

$$\boldsymbol{CORR_{t+1:t+20} = \alpha + \beta_1 DMIDI_t + \beta_2 DMIDI_t^2 + \beta_3 PASTCORR_t + \beta_4 VIXLEVEL_t + \beta_5 PASTRV_t + \varepsilon_t}$$

This specification is used to test whether the relationship between structural drift and subsequent correlation is nonlinear.

In the out-of-sample quadratic specifications, DMIDI is centred using the mean estimated from the training sample before the squared term is constructed. This reduces mechanical dependence between the linear and quadratic regressors without changing the fitted nonlinear relationship.

$$\boldsymbol{DMIDI_C, t = DMIDI_t - bar(DMIDI)_{train}}$$

$$\boldsymbol{DMIDI_{C2}, t = (DMIDI_C, t)^2}$$

Importantly, the centring parameter is estimated using the training sample only. Information from the holdout period is not used in constructing the forecasting regressors.

In the linear specifications, β_1 measures the conditional association between current structural drift and the future target after controlling for conventional market-risk variables. In the quadratic specifications, the marginal relationship depends jointly on the linear and squared DMIDI coefficients.

Regression coefficients are not interpreted as causal effects. The forecasting analysis is designed to assess incremental statistical information and out-of-sample performance rather than to establish causal relationships between structural drift and future market outcomes.

Statistical inference for regressions involving overlapping forward horizons is adjusted using Newey–West heteroskedasticity and autocorrelation consistent standard errors, described in Appendix A.7.

A.7 Newey–West Standard Errors

Several forecasting regressions use forward 20-trading-day outcomes. Because consecutive observations are constructed from overlapping future windows, their regression residuals may be serially correlated. In addition, financial time series frequently exhibit heteroskedasticity.

Conventional OLS standard errors assume that regression errors are uncorrelated and homoskedastic. These assumptions are therefore potentially inappropriate for inference in the present setting, even when the OLS coefficient estimates themselves remain unchanged.

Statistical inference is consequently based on **Newey–West heteroskedasticity and autocorrelation consistent (HAC) standard errors**.

The Newey–West estimator adjusts the estimated covariance matrix of the OLS coefficients for both heteroskedasticity and serial dependence up to a selected lag length.

$$\boldsymbol{Var(\hat{\theta}) = (X^T X)^{-1} \hat{\Omega} (X^T X)^{-1}}$$

where $\hat{\Omega}$ is the HAC estimate of the covariance structure of the regression disturbances.

The Newey–West covariance estimator combines the contemporaneous residual covariance with weighted autocovariances up to lag L.

$$\widehat{\boldsymbol{\Omega}} = \widehat{\boldsymbol{\Gamma}}_{\mathbf{0}} + \sum_{l=1}^{L} \left(\mathbf{1} - \frac{l}{L+1}\right)\left(\widehat{\boldsymbol{\Gamma}}_{l} + \widehat{\boldsymbol{\Gamma}}_{l}^{t}\right)$$

where L is the truncation lag and the term $1 - l/(L+1)$ corresponds to the Bartlett weighting scheme. Autocovariances at shorter lags receive greater weight than those further in the past.

The adjustment is particularly relevant for overlapping forward variables. For example, future realised volatility measured from $t+1$ to $t+20$ shares nineteen return observations with the corresponding target constructed one trading day later. Consecutive forecast errors may therefore exhibit substantial mechanical dependence.

Newey–West corrections affect the estimated standard errors, t-statistics and p-values, but they do not alter the underlying OLS coefficient estimates.

HAC-adjusted inference is used when assessing the statistical significance of the predictive relationships examined in the project. This is particularly important for the nonlinear DMIDI specifications, where apparently significant historical relationships must be distinguished from results that remain statistically credible after accounting for time-series dependence.

Statistical significance is nevertheless treated as only one dimension of model evaluation. The project places greater emphasis on genuine out-of-sample forecast performance, which is described in Appendix A.8.

A.8 Out-of-Sample Evaluation Methodology

Out-of-sample evaluation is used to determine whether the predictive relationships identified in the historical sample remain useful when applied to previously unseen observations. This distinction is essential because good in-sample fit does not necessarily imply genuine forecasting ability.

The project uses both a fixed holdout design and an expanding-window validation procedure.

Fixed training and holdout split

In the primary holdout design, observations from **2015 to 2021** are used for model estimation, while the period from **2022 to September 2026** is reserved for out-of-sample evaluation.

Model coefficients, centring parameters and all other quantities required for prediction are estimated using the training sample only. No information from the holdout period is used during model fitting.

RMSE

Forecast accuracy is evaluated using Root Mean Squared Error (RMSE):

$$\boldsymbol{RMSE} = \sqrt{\left(\frac{\mathbf{1}}{\boldsymbol{T}}\sum_{t=1}^{T}(\boldsymbol{y_t} - \widehat{\boldsymbol{y}}_{\boldsymbol{t}})^{\mathbf{2}}\right)}$$

where y_t is the realised target and $\hat{y}_t$ is the corresponding model forecast. Lower RMSE values indicate better predictive accuracy.

MAE

Mean Absolute Error (MAE) is also reported as a complementary measure:

$$\boldsymbol{MAE} = \frac{\mathbf{1}}{\boldsymbol{T}}\sum_{t=1}^{T}|\boldsymbol{y_t} - \widehat{\boldsymbol{y}}_{\boldsymbol{t}}|$$

MAE is less sensitive than RMSE to very large forecast errors and therefore provides a complementary view of average predictive performance.

Out-of-sample R^2

Model performance relative to a benchmark is summarised using the out-of-sample R^2:

$$R^2_{OOS} = 1 - \frac{MSE_{model}}{MSE_{benchmark}}$$

A positive OOS R^2 indicates that the evaluated model produces lower mean squared forecast error than the benchmark. A negative value indicates that the benchmark performs better.

Expanding-window validation

To assess whether the results depend excessively on a single train-test split, the project also uses expanding-window validation. For each evaluation year, the model is re-estimated using all eligible observations available prior to that year.

The estimation sample therefore expands through time while the evaluation period remains strictly subsequent to the data used for model fitting.

This procedure produces a sequence of genuinely out-of-sample forecasts across multiple years and makes it possible to assess whether any forecasting improvement is recurrent or concentrated in a single episode.

Purge gap

Because the dependent variables are constructed over forward 20-trading-day horizons, a **20-trading-day purge** is imposed around the boundary between estimation and evaluation samples. This reduces the risk that overlapping forward windows create information leakage between the training target and the holdout period.

Out-of-sample evaluation is given greater weight than in-sample statistical significance when assessing the usefulness of Market DNA as a forecasting tool. A relationship may be statistically significant within the estimation sample yet fail to improve genuine forecast accuracy.

For this reason, the project distinguishes explicitly between **historical association**, **statistical significance** and **out-of-sample predictive performance**.

A.9 Monte Carlo Methodology

The Monte Carlo application translates Market DNA states into empirical portfolio-risk scenarios using a historical block-bootstrap procedure. Rather than imposing a parametric distribution on asset returns, the simulation resamples historical multi-asset return blocks observed under specific structural market conditions.

Three scenario sets are considered: Stable / Local states, Current-like analogues and Systemic Structural Shifts.

Scenario-conditioned block selection

Historical signal dates are assigned to scenario sets using the Market DNA regime framework. Current-like analogues are selected according to proximity to the current observation in the two-dimensional space defined by DMIDI percentile and Drift Breadth percentile.

For each eligible signal date t, the subsequent five trading days of six-asset returns are retained as one empirical return block.

$$B_t = \{R_{t+1}, R_{t+2}, \dots, R_{t+5}\}$$

where each R_t is the vector of contemporaneous returns for QQQ, XLF, XLE, FEZ, TLT and GLD.

Non-overlapping blocks

To reduce duplication of closely related historical information, the final specification uses **non-overlapping five-day blocks** within each scenario set.

This filtering produces 133 blocks for Stable / Local states, 48 blocks for Current-like analogues and 136 blocks for Systemic Shift states.

Non-overlapping blocks are preferred for the final results because overlapping forward windows may repeatedly reuse the same underlying market observations and artificially increase the effective sample size.

Simulation horizon

Each simulated portfolio path spans **20 trading days** and is constructed by randomly drawing four five-day historical blocks, with replacement, from the relevant scenario set.

A total of **10,000 simulated paths** is generated independently for each scenario.

Portfolio construction

The simulated portfolio is equally weighted across the six assets. Let $r_{i,t}$ denote the simple return of asset i on day t. The daily portfolio return is:

$$r_{p,t} = \frac{1}{N}\sum_{i=1}^{N} r_{i,t} \quad \text{with N} = 6$$

Equal weighting is used to keep the risk application transparent and to avoid introducing an additional portfolio-optimisation layer.

Cumulative path construction

Simulated cumulative portfolio wealth evolves according to:

$$V_t = V_{(t-1)(1+r_{p,t})} \quad with\ V_0 = 1$$

The 20-day terminal portfolio return is calculated from the final simulated wealth level, while maximum drawdown is computed from the full intra-path wealth trajectory.

Why block bootstrap?

The block-bootstrap approach preserves two forms of empirical dependence that would be lost under independent daily resampling. First, all six asset returns are retained jointly within each day, preserving contemporaneous cross-asset dependence. Second, the five-day block structure preserves short-run temporal dependence within each sampled episode.

The procedure therefore generates scenario-conditioned empirical distributions while remaining directly linked to historically observed return dynamics.

The Monte Carlo exercise is not intended to estimate an unconditional probability distribution for future returns. Its purpose is to compare empirical portfolio-risk distributions conditional on historically observed Market DNA states.

Results should therefore be interpreted as **scenario-conditioned historical simulations rather than forecasts**. The quality of each scenario distribution depends on the number, diversity and representativeness of historical blocks available within the corresponding Market DNA state.

A.10 Additional Robustness Results

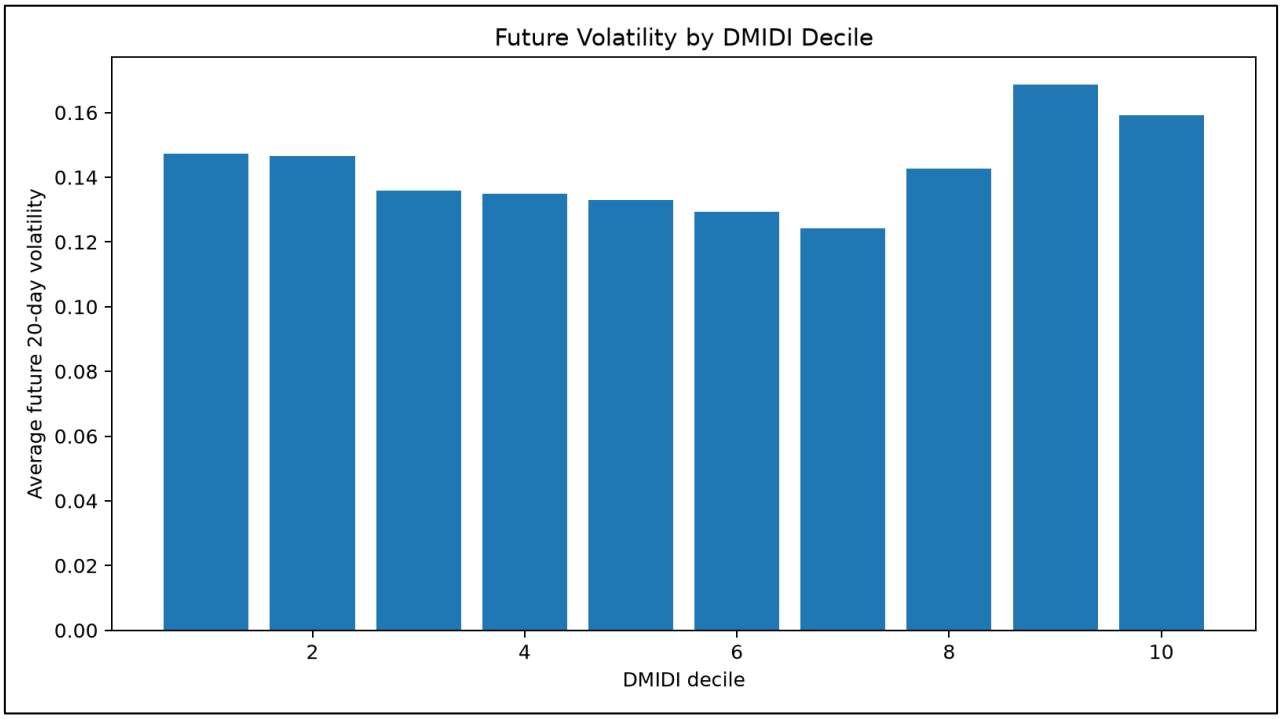
Additional robustness checks are reported to assess whether the main conclusions depend excessively on individual modelling choices, particular historical episodes or specific visual patterns. These results are presented separately from the main analysis because they support interpretation rather than materially change the central conclusions of the project.

Robustness of the volatility result

The relationship between DMIDI and future realised volatility was also examined across DMIDI deciles. The descriptive pattern is nonlinear, with some higher-drift groups exhibiting elevated subsequent volatility. However, the pattern is not monotonic and does not survive the inclusion of conventional volatility controls in the regression analysis.

In particular, neither the linear nor the squared DMIDI term is statistically significant in the controlled quadratic volatility specification. The decile results should therefore be interpreted as exploratory descriptive evidence rather than as evidence of a robust forecasting relationship.

Figure A1. Future Realised Volatility Across DMIDI Deciles



Note: The figure reports descriptive future realised volatility across DMIDI deciles. The observed nonlinear pattern does not remain statistically significant after controlling for the VIX level and past realised volatility.

Decile-based analysis produces similar descriptive nonlinearities, but these patterns should be treated as exploratory. The subsequent out-of-sample tests provide the more relevant assessment of whether the nonlinear relationship contains genuine predictive information.

Monte Carlo robustness: overlapping versus non-overlapping blocks

The initial Monte Carlo implementation used overlapping five-day forward return blocks. Because adjacent signal dates can generate highly similar return sequences, the simulation was repeated using non-overlapping historical blocks.

The number of available blocks falls materially under the non-overlapping specification, but the principal qualitative conclusions remain similar. The Stable / Local scenario continues to exhibit the largest extreme-tail losses, while the Systemic Shift scenario remains intermediate in terms of severe downside risk.

Table A2. Monte Carlo Tail-Risk Robustness

Scenario	VaR 99%	ES 99%
Stable / Local	12.81%	14.24%
Current-like	6.63%	7.74%
Systemic Shift	8.61%	10.35%

Note: Results correspond to the final non-overlapping-block specification. The qualitative ordering of extreme tail risk is consistent with the robustness analysis performed using alternative block construction.

Current-like analogue concentration

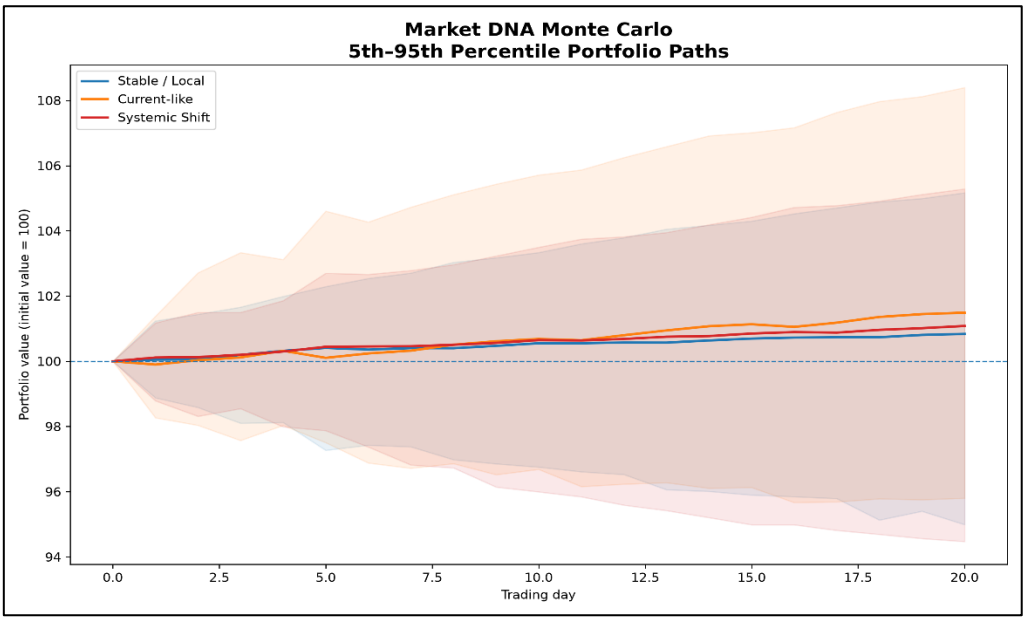
A further robustness issue concerns the historical diversity of the Current-like scenario. The original set of 250 nearest Market DNA analogues is highly concentrated in a limited number of years.

Approximately **54.8%** of the observations originate from 2026 and **23.6%** from 2020. The remaining observations are distributed across only a small number of additional years. The effective number of equally represented historical years is therefore approximately **2.7**.

This concentration limits the independence and historical diversity of the Current-like scenario. Consequently, its relatively favourable terminal-return and extreme-tail results should not be interpreted as evidence that the current structural regime is intrinsically low risk.

Portfolio-path robustness

Figure A2. Simulated Portfolio Paths by Market DNA Scenario



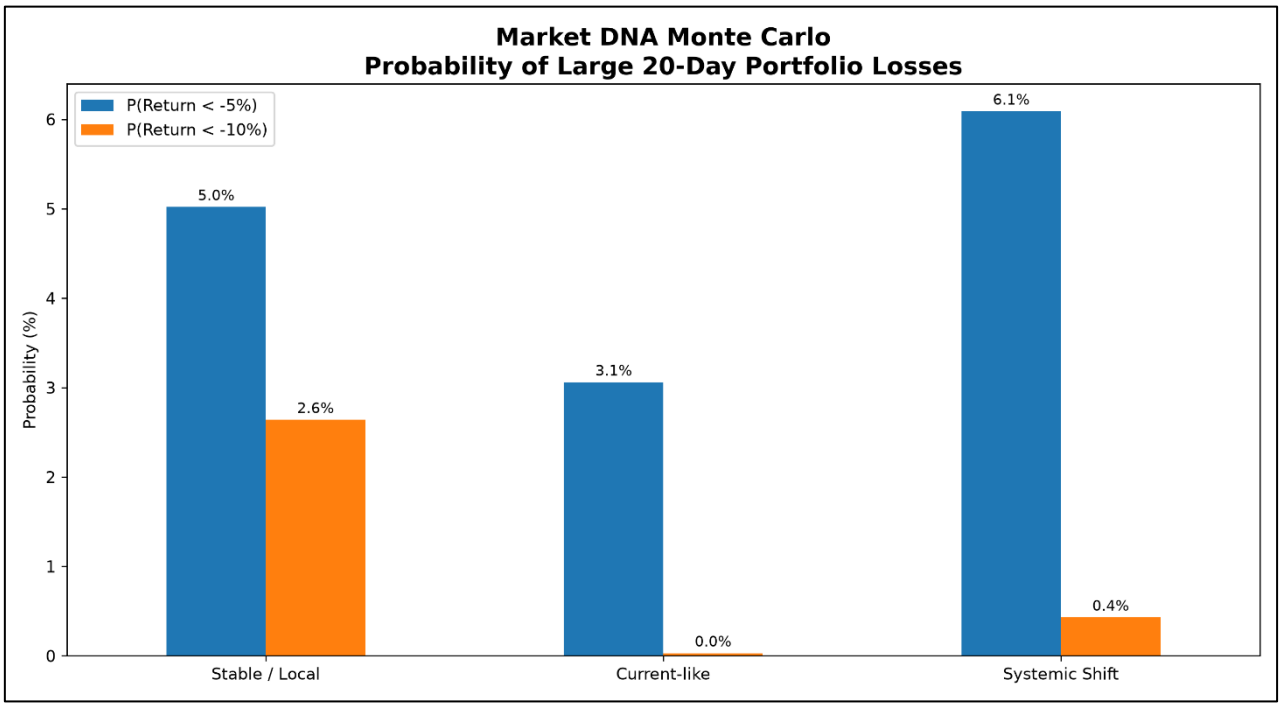
Note: The figure displays simulated 20-day portfolio paths and their empirical uncertainty bands under each Market DNA scenario. Path dispersion reflects scenario-conditioned historical return dynamics rather than a parametric return distribution.

The path simulations highlight a distinction that is less visible in terminal-return statistics. Although the Current-like scenario exhibits comparatively favourable terminal tail metrics, its simulated paths can still

experience meaningful temporary declines. This is consistent with its relatively large average maximum drawdown.

Loss probabilities

Figure A3. Simulated Portfolio Loss Probabilities



Note: Probabilities are calculated from the simulated 20-day terminal portfolio returns. The 5% and 10% loss thresholds provide frequency-based complements to VaR and Expected Shortfall.

The loss-probability results reinforce the non-monotonic relationship between Market DNA regimes and portfolio risk. Systemic Shift states exhibit the highest probability of a loss greater than 5%, while Stable / Local states contain the greatest probability of more extreme losses beyond 10%.

Taken together, the robustness exercises do not alter the central conclusions of the project. Market DNA does not provide robust incremental information for forecasting future volatility, while its relationship with future cross-asset correlation remains modest and nonlinear. The Monte Carlo results are also sensitive to the historical composition of the scenario sets, although the principal qualitative findings survive the use of non-overlapping return blocks.

These results reinforce the interpretation of Market DNA as a **structural monitoring and scenario-conditioning framework rather than a standalone forecasting system**.